

Grand Responsive Universe Theory

*The Emergence of Everything: A Candidate Theory of
Everything Organized Around the Emergence of Responsiveness*

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*121 registered claims · six tiers, machine-checked · 28 open negatives documented
Every claim is marked at the tier it earned. Every negative is documented. Nothing is fitted away.*

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Abstract. This is the public record of the v4 program of *GRUT* — Grand Responsive Universe Theory. Its premise is one sentence: the gravitational vacuum is not an inert stage but a *responsive medium with finite memory*. Its thesis is one more — everything downstream follows from a single transition: the moment the universe became able to hold its own past. Before that moment there is reaction without record; after it, the medium can *remember*, and with memory come persistence, an arrow of time, the possibility of structure, and, far up the chain, observers who turn back to study the responsiveness they are themselves made of. This book runs that history forward in the order the universe built it — **Q** → **F(t)** → **Responsiveness** → **Vacuum** → **Physics** → **Complexity** → **Observation** — from the pre-responsive substrate, through the switching-on of memory as the cosmos cools, to the physics it makes possible, the structure that physics builds, and the mind that structure eventually produces. Each rung is placed at its honest tier — **DERIVED** from GRUT's own structure, **HOSTED** as a received input, **FORBIDDEN-BY-THEOREM** from its own axioms, **OPEN** but attackable, or **CONJECTURAL** — and a decisive *organize-or-host* test draws the boundary of the theory's reach: dark energy, the mass hierarchy's existence, α , and dark matter (resident in the substrate $F(t)$) organize around responsiveness; quantum gravity is half-derived (response yes, geometry hosted); the Standard-Model matter content is hosted; the observer and meta frontiers remain conjectural. This is a **candidate framework** — **the current rung, not the arrival**: a complete tiered *map* with a sharp edge, not a derivation of everything.

How this was reached — **GRUT-RAI**. Every quantitative claim in this book is carried by **GRUT-RAI** (GRUT Responsive AI), the open research platform on which the theory was built and stress-tested. GRUT-RAI is not prose *about* physics — it is the physics, as runnable code: each claim is a structured record binding the statement to the function that computes it, the tests that gate it, the tier it has earned, and the single observation that would falsify it (**121 claims, six machine-checked tiers, 3,000+ gating tests**). The framework was developed and corrected through *adversarial verification* — independent AI agents recompute every load-bearing number, re-derive every result, and try to refute every claim; a claim rises a tier only when a test certifies it, and is demoted and logged the moment one fails. What follows is what survived that process — including the places it forced GRUT to retract its own over-claims. The platform, the registry, and every line of derivation code are open:

[*github.com/ryangrvt/GRUT-RAI*](https://github.com/ryangrvt/GRUT-RAI)

The Thesis

The universe is a sequence of emergent layers.

Responsiveness is the transition that turns a passive universe into a participating one — a universe that can hold its own past. Before it, there is reaction without record: events happen and leave nothing behind. After it, the medium can *remember* — and with memory comes persistence, an arrow of time, the possibility of structure, and, far up the ladder, observation and self-reflection. Everything this book describes is downstream of that one transition.

The purpose of the book is to **run that process forward** — from the earliest substrate, through the switching-on of responsiveness, to the physics it makes possible, the structure that physics builds, and the observers that structure eventually produces: observers who turn back to study the responsiveness they are themselves made of.

In one line — the universe's own order of emergence, which is also the order of this book:

$Q \rightarrow F(t) \rightarrow \text{Responsiveness} \rightarrow \text{Vacuum} \rightarrow \text{Physics} \rightarrow \text{Complexity} \rightarrow \text{Observation} \rightarrow \text{the universe}$ discovers responsiveness.

Before the chapters open — the physics in one look. GRUT has a single engine: one *closed-time-path* (CTP, or "in-in") action — the Schwinger–Keldysh formulation in which physics is the response not to a state but to *realized differences* between a history and itself. The forward branch of history is compared against the backward branch, and only what actually happened — the difference on the closed contour — sources a response. This is the pillar **Q**: causal, unitary, with the arrow of time built in *before any clock exists*. It is the framework's single strongest piece — a theorem of the formalism, not an assumption.

Vary that action once and a single **constitutive equation** falls out — the load-bearing hub every later sector is a regime of, with its single-pole susceptibility:

$$\tau_0 \dot{Z} + Z = Z_{\text{target}}, \quad \chi(\omega) = \frac{\alpha}{1 - i\omega\tau_0}$$

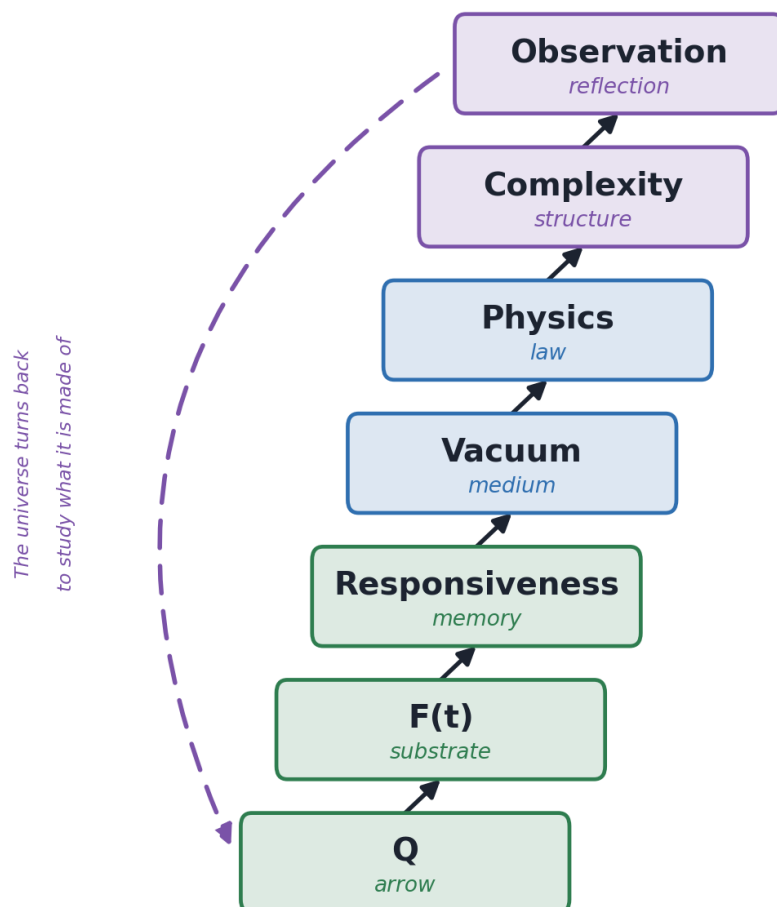
— a relaxation law in which the responsive vacuum chases its target with one finite memory time, its response a single pole in the frequency plane. Two constants fix it, and nothing else does: a macroscopic **relaxation time** $\tau_0 \approx 41.9 \text{ Myr}$ (the pillar **F** — finite memory, the one dimensionful input the theory must be told) and a dimensionless **impedance** $\alpha = 1/3$. From these two numbers follow the framework's signature export — the deep-infrared gravitational refractive index $\mathbf{R} = \sqrt{4/3} \approx 1.1547$ — and dark energy as a *terminal velocity* of the relaxing vacuum ($\Omega_\Lambda \approx 0.6886$). There are no adjustable knobs in the predictive core.

Its two famous limits are not added by hand; they are the two faces of the one law. In the **fast limit** (memory negligible, $\tau \rightarrow 0$) the constitutive update is the Schrödinger equation — quantum mechanics recovered, not postulated. In the **slow, sub-horizon limit** it is general relativity. Read structurally, GRUT is general relativity's long-wavelength scale-rescaling redundancy (the pillar **D**), broken in a controlled way by exactly one proper length, $\mathbf{L}_0 = \mathbf{c} \cdot \tau_0 \approx 12.85 \text{ Mpc}$ — the same shape by which a particle's mass breaks scale invariance. A theory is defined by which symmetry it breaks, and by how much; this is GRUT's. With the engine in view, the rest of the book runs it forward.

The Emergence Ladder

the universe's own order of emergence

→ the universe discovers responsiveness



GRUT ToE v4 — The Emergence of Everything | D. Ryan Grover | $Q \rightarrow F(t) \rightarrow \text{Responsiveness} \rightarrow \text{Vacuum} \rightarrow \text{Physics} \rightarrow \text{Complexity} \rightarrow \text{Observation}$

Figure 1. The emergence ladder — the universe's own order of emergence.

What follows is not a cosmology book, a gravity book, or a consciousness book. It is **a developmental history of reality, organized around the emergence of responsiveness**. The discipline that keeps that history honest — what is derived, what is hosted, what is forbidden, what remains open — sits *beneath* the story, in the next section, where a referee needs it. It does not lead, because it is the scaffolding, not the building.

How to Read This — The Discipline Beneath the Story

The thesis above is the affirmative claim. This section is the discipline that keeps it honest: the distinctions a referee needs, positioned under the narrative rather than in front of it.

This document runs the universe forward — from before there was time, to the observers who would one day measure it — and places each thing that emerges at its *true tier*. It is the v4 **spine**: a single developmental narrative, ordered the way the universe itself emerged rather than the way a physics textbook is organized. It first poses the questions the framework exists to answer, then states what GRUT is, then tells the story of emergence rung by rung, audits the open frontiers under a strict format, and finally asks what kind of theory the whole thing amounts to.

Two distinctions govern everything that follows.

Umbrella vs. deliverable. *GRUT* — Grand Responsive Universe Theory — is the umbrella principle that the gravitational vacuum is a *responsive medium* rather than an inert stage. It is not, by itself, a finished theory of everything; it is the stance from which the finished theories are built. The falsifiable rungs built on that stance are named *GRUT ToE vN*. This document is the v4 spine of that program. Where the prose says "GRUT derives," read it as "the deliverable, built on the umbrella principle, derives."

What makes it a Theory of Everything — the destination, not the present state. First, a tense correction this document holds to throughout: **GRUT ToE v4 is the current rung, not the arrival.** *GRUT* (the umbrella) is the program; the **GRUT ToE vN** are its falsifiable rungs; **a complete GRUT ToE is the result the program walks toward** as the open rungs close. Aiming at that destination is what any serious program is entitled to do; asserting one has arrived is not. So nothing below should be read as "v4 is the finished theory of everything" — it is the latest honest map.

Second, what *kind* of completeness the destination has. A ToE, in the sense meant here, is **not** a theory that *derives* everything. It is a theory that has a coherent, tiered **place** for everything, with a **sharp boundary** — completeness of the *map*, not omniscience. And this is a **deliberate commitment, not a shortfall**: GRUT's defining minimalism (one dynamical metric, one memory channel, Q) *forbids* the moves a derive-everything ToE would require, so the completed GRUT ToE is a complete tiered **map** by construction. The places where GRUT must *receive* an input (HOSTED) or *forbids* something from its own axioms (FORBIDDEN) are not gaps in the story. They are the **seams** of a layered ontology showing through, woven into the narrative as such: *here the universe makes X; here it must receive Y; here it forbids Z.*

Third, adaptability — stated precisely so it is not a loophole. Some walls are **postulates** (single-mode is a channel-counting choice) and genuinely move with more work; those OPEN cells are the rungs. Others are **theorems** (locality/no-halo; the propagating-relic pincer) and move only by breaking a premise — Q or locality — which does not "adapt" the theory but **constitutes a different theory, not GRUT**. A separate move stays inside GRUT: descending into a **deeper substrate layer of GRUT** (the bath F(t) it already posits *beneath* the vacuum, where those theorems have no jurisdiction). That descent breaks no premise — it is not a successor theory; F(t) is a level GRUT's own ontology already contains (Q → F(t) → responsiveness → vacuum → physics), so giving it field content leaves Q, locality, the vacuum sector, and the theorem-walls all intact; it is simply conjectural and below the zero-parameter v4 core. The GRUT umbrella encompasses that downward growth, so "complete" can *deepen* across versions; but each deliverable vN states honestly which cells it derives, hosts, and

forbids *today*. Anti-salesmanship is the point: a spine that quietly upgrades a HOSTED or OPEN result to DERIVED is the one failure mode that sinks the whole thing on contact with a referee, and this document refuses that upgrade everywhere it is tempting.

The Boundary Legend (Five Tiers)

The Boundary Legend — Five Tiers		
Every emergence event carries exactly one tier; the honesty is the boundary canonical example		
	DERIVED GRUT makes it from its own structure (zero / low free parameters).	$\mu_{\text{linear}} = 1$ — the P^{TT} projector theorem ($\sim 32\sigma$ ISW)
	HOSTED GRUT receives it as a boundary condition it does not generate.	Dark matter — substrate-resident in $F(t)$, not external
	FORBIDDEN-BY-THEOREM GRUT rules it out from its own axioms.	Hierarchy magnitude — a 2nd propagating vacuum pole forbidden
	OPEN No mechanism yet, but attackable.	The $\alpha = 1/3$ antecedent — the IR-carrier identification
	CONJECTURAL A structural hook only — far from derivation.	Consciousness — the in-in self-comparison hook, not a theory of mind
Several events are split-tier seams [SPLIT]: a mechanism DERIVED on an anchored input — the split is the honest content, never collapsed upward.		

GRUT ToE v4 reader legend (5 tiers) over the machine registry (6 tiers, 121 claims) — GRUT_V4_SPINE.md

Figure 2. The five-tier boundary legend — every emergence event carried at its earned tier.

Every emergence event below carries exactly one of these five reader-facing tags. They are the *narrative* legend. Beneath them sit the **six machine-checked registry tiers** in `grut/toe/registry.py` (computed / anchored / conjectural / open_negative / foundational / meta); the narrative tiers map onto the machine tiers as: **DERIVED** $\approx$ computed or foundational; **HOSTED** $\approx$ anchored; **OPEN** $\approx$ open_negative; **CONJECTURAL** $\approx$ conjectural; **FORBIDDEN-BY-THEOREM** is a narrative reading of computed/conjectural no-go claims; and **meta** is the audit layer (the registry checking itself), which carries no reader-facing tier of its own. There are not two competing tier systems — there is one machine layer of six tiers and one reader legend of five, related by this map.

Tier	Meaning	Canonical example
DERIVED	GRUT makes it from its own structure (zero / low free parameters).	$\mu_{\text{linear}} = 1$ (the projector theorem); Koide $K = 2/3$ as a Z_3 identity; the arrow of time from $\dot{S} \geq 0$; QM in the $\tau \rightarrow 0$ limit; Q, the in-in causal arrow.
HOSTED	GRUT <i>receives</i> it as a boundary condition it does not generate.	Dark matter — hosted by the locality/no-halo theorem plus the single-mode channel-counting postulate. Also the SM gauge group; τ_0 and τ_{micro} as anchors.

Tier	Meaning	Canonical example
FORBIDDEN-BY-THEOREM	GRUT <i>rules it out</i> from its own axioms.	The hierarchy magnitude; a propagating dark relic / second vacuum pole; linear-scalar enhancement $\mu_{\text{linear}} \rightarrow 4/3$ (refuted at $\sim 32\sigma$).
OPEN	No mechanism yet, but attackable.	The $\alpha = 1/3$ antecedent (the Riegert / IR-carrier identification); the D initial-state condition; baryogenesis magnitude; the Born-rule weights; the flavor mechanism.
CONJECTURAL	A structural hook only — far from derivation.	Consciousness — the in-in self-comparison is a hook, <i>not</i> a theory of mind. Also the black-hole interior / "1 Space"; the $\theta = 2/9$ candidate identity.

Several events are **split-tier seams**: the mechanism is DERIVED but its input is anchored, or a *form* is DERIVED while its *antecedent* is OPEN. These are tagged [SPLIT], and the split is the honest content — it must not be collapsed to the higher tier.

The Through-Line, in Mechanism

The thesis states the ladder as one line; this is the mechanism it names. A closed, self-referential universe in which the always-present causal arrow (Q) condenses a pre-responsive substrate $F(t)$ into a single memory channel (responsiveness), and with it a responsive vacuum, the known physical sectors, cosmic structure, and finally the observers that are themselves crystallized medium — observers who turn back to study the responsiveness they are made of. The no-gos are carried not as embarrassments but as the load-bearing seams that make the map complete and the boundary sharp.

Part Map

The body is a five-part architecture, ordered chronologically/developmentally — the sequence in which the universe itself built each thing, not the order of a physics syllabus.

- **Part I — The Discovery of Responsiveness** (Ch 1–3): the floor (Q), the substrate $F(t)$, and the emergence event (responsiveness), before any physics exists.
- **Part II — Emergence of Physics** (Ch 4–7): the responsive vacuum, then the known sectors — quantum reality, gravity, cosmos — as regimes of one constitutive law.
- **Part III — Emergence of Complexity** (Ch 8–11): structure, life, mind, intelligence as honestly-tiered *placed slots* on the ladder, not GRUT derivations.
- **Part IV — The Open ToE Frontier** (eight frontiers): each posed as "How Does X Emerge in a Responsive Universe?" and answered under a strict five-point format.
- **Part V — The Emergence of Everything** (Ch 12–14): the ladder retold as one narrative, why the name is earned, and the decisive ongoing test.

Appendix A carries the overclaim-risk table; Appendix B carries the verified numbers.

PART I — THE DISCOVERY OF RESPONSIVENESS

$Q \rightarrow F(t) \rightarrow \textbf{Responsiveness} \rightarrow \textit{Vacuum} \rightarrow \textit{Physics} \rightarrow \textit{Complexity} \rightarrow \textit{Observation} \rightarrow (\textit{the universe discovers responsiveness}).$

This Part covers the three deepest rungs of the spine: **$Q \rightarrow F(t) \rightarrow \textbf{Responsiveness}$** . It is the floor of the whole ontology, told as the universe's own first moments. Chapter 1 poses the questions the framework exists to answer. Chapter 2 lays out the framework in a single view — Q , $F(t)$, responsiveness, the constitutive law, the layered ontology, and the tier legend in action on every pillar. Chapter 3 runs the clock back to before there was time, and forward again through the one event that started it: the emergence of responsiveness. No physics — no quantum mechanics, no gravity, no cosmos — exists yet when this Part begins. Those are Part II. Here we establish only the floor (Q), the substrate ($F(t)$), and the emergence event that turns the one into the other.

Chapter 1 — The Problem

Before any equation, a theory of everything owes an account of a handful of facts so familiar that physics usually declines to explain them and merely assumes them. GRUT's claim to the title rests less on the answers it can derive — those are tiered, frontier by frontier, in the chapters that follow — than on the *shape* of the questions it takes as primary. So this chapter only poses them. It asserts no derivation; the later chapters do the tiering.

Why does anything remember? The microscopic laws are time-reversible, yet the universe at large keeps a record. Disturbances leave traces; the past is fixed and the future is not. A medium that merely *responded* — a perfect spring that snaps back the instant the load is removed — would carry no history. Something has to hold the displacement for a while. The question is not "why is there friction" but the deeper one beneath it: why is there a *finite duration* over which the world holds the shape of what happened to it, rather than zero (perfect elasticity, no memory) or infinity (perfect rigidity, no relaxation)?

Why is there an arrow of time? The same time-symmetric equations are run forward in every textbook and backward in none. The Second Law is usually imported as a statistical boundary condition on initial states. But a closed universe has no exterior to set its boundary conditions from, and no observer outside it to declare which way is "forward." Where, *internal to the structure*, does the asymmetry between past and future come from — and does it come *before* or *after* the universe has a clock?

Why does information survive? Quantum evolution is unitary: information is redistributed, never destroyed. Yet the world looks irreversible, definite, classical at large scales. How does a unitary universe produce a stable record of itself — one that can be stored, copied, and eventually read — without violating the conservation that makes it unitary in the first place?

Why does a universe have a history at all? Take the three together and they compose into one question. A history is a remembered sequence with a direction and a preserved record. Memory, arrow, and information-survival are not three problems but three faces of one: *what does it take for a universe to have a past it carries forward?*

GRUT's wager is that all four reduce to a single structural fact, and that this fact is logically *prior* to time itself. The fact is **Q** — the principle that the vacuum responds to realized, physically distinguishable differences from its own past, and never to its future. On the closed-time-path contour that governs a closed quantum universe, the influence action vanishes identically when the forward and backward branches of history agree, $S_{IF}[\phi_+ = \phi_-] = 0$; physical response is the variation with respect to their *difference*. This is not yet a clock — **Q** contains no timescale — but it is already a *direction*: response-to-past, never-to-future, before any memory length exists.

The chapter's only claim, then, is a claim about ordering, and it is the one the rest of Part I makes good on: the arrow of time does not arrive *with* memory or *with* the responsive vacuum. It **predates** them. Memory, the responsive vacuum, structure, and observers are what a universe builds *on top of* an arrow it had from the start. Everything after this chapter is the construction of that "on top of," placed at its honest tier. (Motivational; no tier of its own — this chapter poses what later chapters answer.)

Chapter 2 — GRUT in One View

This chapter is the legend the rest of the document is read against. It is not yet narrative; it is the standing description of the framework — Q , $F(t)$, responsiveness, the constitutive law, the layered ontology — read against the five-tier boundary legend so that every later emergence event can be tiered honestly rather than asserted.

GRUT begins from one ontological commitment: **the gravitational vacuum is a medium that responds, not an inert background.** A medium that responds has a response function, and a response function has structure — and the content of the theory comes from the structure, not from the commitment. That structure rests on four inputs of very unequal standing, exactly one of which is a true zero-input theorem.

The Four Inputs

Q — the in-in causal arrow. — DERIVED. The Schwinger–Keldysh closed-time-path (CTP) action describes the universe by two copies of its own history, a forward branch ϕ_+ and a backward branch ϕ_- , with physical response defined as the variation of the influence functional S_{IF} with respect to their difference. Q is the set of structural facts that follow: the closed–closed propagator is identically zero; the response on the diagonal ($\phi_+ = \phi_-$) is exactly zero; and the fluctuation–dissipation / KMS relation ties the noise N to $\text{Im } \chi$ with $N \geq 0$. Q contains no τ_0 , no τ_{micro} , no α — it is scale-free — and it supplies a response-to-past, never-to-future directionality *before any memory length exists*. This is GRUT's single strongest piece: a theorem of the formalism, verified on a concrete driven-oscillator example in all four legs (field doubling invertible, variation principle reproducing the equation of motion, retarded propagator causal, FDT recovering both classical and quantum limits). It is the only true zero-input SOLID node in the graph, and the highest-fanout claim in the registry — 93 downstream dependencies. (`ctp_action_structure`, *tier computed*, `registry.py:451`; `grut/foundation/ctp_action.py`, *verify 5/5.*)

F — finite single-pole memory. — HOSTED [SPLIT]. The vacuum relaxes on *one* time $\tau_0 \approx 41.9$ Myr, giving a susceptibility $\chi(\omega) = \alpha/(1 - i\omega\tau_0)$: a single pole at $\omega = -i/\tau_0$ in the lower half-plane (causal, Kramers–Kronig compatible, recovering GR as $\omega \rightarrow \infty$). Equivalently, the vacuum obeys the constitutive law $\tau_0 \dot{z} + z = z_{\text{target}}$, equivalently the exponential kernel $K(t) = (1/\tau_0)e^{-t/\tau_0}\Theta(t)$ — one object in three representations. The single-pole *form* is not arbitrary: it is the Markovian limit of a Mori–Zwanzig projection of the slow transverse-traceless shear variable, exact to first order in τ_K/τ_0 , and an off-axis "dark-capable" pole would require $\tau_K > \tau_0/4$, which the 34-order scale separation forbids. **But the form being derived is not the same as the pillar being derived. F is POSTULATED, and the value τ_0 is anchored — Q generates neither.** Q alone does *not* force single-pole-ness: passivity (Herglotz) is strictly weaker than complete monotonicity, and a passive kernel can carry negative Debye weight or off-axis poles — the registry says so in plain words, "*single-pole-ness is NOT a theorem of Q alone*" (`registry.py:5287`). So the *form* is DERIVED (the MZ Markovian limit), while F as a pillar and the value of τ_0 are taken in, not made. `*(finite_memory_form_from_q, computed, registry.py:5278` — covers the single-pole form via MZ; `first_order_from_mori_zwanzig, computed, registry.py:5422`; `tau_0_derivation, registry tier computed at registry.py:180` — its statement POSITS τ_0 with two independent anchors, i.e. the *value* is anchored even though the claim's tier field is "computed"; `mori_zwanzig_kernel.py`, *verify 8/8.*)*

OVERCLAIM FLAG (Appendix A #20, #6). Do not write "*F is derived from Q.*" The single-pole form is the MZ Markovian limit (derived); *F-as-pillar* is POSTULATED and the τ_0 value is anchored. FDT-consistency with Q is consistency, not derivation.

D — the broken dilatation redundancy. — OPEN [SPLIT]. A scale-free theory carries an adiabatic spatial-dilatation redundancy — the Weinberg adiabatic mode, a large/residual diffeomorphism, an exact gauge redundancy in the $L_0 \rightarrow 0$ limit. The arrival of a finite memory time gives the vacuum exactly one proper length, $L_0 = c\tau_0 \approx 12.85$ Mpc, and a fixed proper length is logically incompatible with that redundancy. The breaking enters at $O((L_0/k_{\text{phys}})^2)$; it is a diffeomorphism, not a Weyl rescaling, so the measure Jacobian is identically one and α does not enter; and it is confined to the tensor sector — which is exactly why linear scalar cosmology is left untouched. The *breaking* is rigorously computed and the Keldysh-doubling content proven; what is *presupposed* is the $L_0 \rightarrow 0$ redundancy itself, imported from Weinberg rather than re-derived, with one undischarged residue (the scale-free in-in initial-state condition, GAP-1). A pure redundancy cannot be spontaneously broken; what L_0 breaks is its *boundary charge*. (*adiabatic_dilatation_redundancy_nogo*, *computed*, *registry.py:5227*; GRUT_V3_ORGANIZING_STRUCTURE.md *Bridge D*.)

$\alpha = 1/3$ — the adopted dimensionless axiom. — OPEN [SPLIT]. The single dimensionless input, adopted as c and $\hbar$ are adopted. It sets the DC response amplitude $\chi(0) = \alpha$, fixes the deep-IR refractive index $R = \sqrt{4/3} = 1.15470$, the screening factor $S = 12\pi/\alpha^2 = 108\pi = 339.292$, and the response normalization. A *conditional* theorem is verified Fraction-exact: if the gravitational conformal mode is the IR carrier of the vacuum susceptibility, then the trace-anomaly ratio $a/c = 1/3$ (Duff; Komargodski–Schwimmer). But the antecedent is unproven and the value is adopted — this is the single most load-bearing open gap in the framework, fanning out maximally: α 's leverage lives in the screening $S = 12\pi/\alpha^2 \propto 1/\alpha^2$ — a ± 0.07 shift in α moves the screened Hubble scale $H_0 = 1/(S\tau_0) \propto \alpha^2$ by roughly a factor of ~ 2 (.venv check: S runs $558 \rightarrow 339 \rightarrow 236$ across $\alpha = 0.26 \rightarrow 1/3 \rightarrow 0.40$), while the tree dark-energy fraction $(2-R)^2$ is screening-free and shifts only $\sim 15\%$ ($0.770 \rightarrow 0.714 \rightarrow 0.667$). The value's standing is decisive. (*alpha_vac_axiom*, *foundational*, *registry.py:238*; *alpha_vac_derivation*, *open_negative*, *registry.py:274*; *conformal_mode_scalar.py*, $a/c = \text{Fraction}(1,3)$, *verify 9/9*.)

Spec — $\alpha = 1/3$, the single adopted axiom. Tier: OPEN [SPLIT]. The one dimensionless input, adopted as c and $\hbar$ are. It sets the DC response $\chi(0) = \alpha$, fixes $R = \sqrt{4/3} = 1.15470$ and the screening $S = 12\pi/\alpha^2 = 108\pi = 339.292$. How (the split): a conditional theorem is proven Fraction-exact — if the gravitational conformal mode is the IR carrier of the susceptibility, then the trace-anomaly ratio $a/c = 1/3$ (Duff; Komargodski–Schwimmer) — but the antecedent is unproven and the value is **adopted, not derived**. This is the most load-bearing open gap: a ± 0.07 shift in α moves the screened scale $H_0 = 1/(S\tau_0) \propto \alpha^2$ by $\sim 2\times$ through $S = 12\pi/\alpha^2$, while the tree $(2-R)^2$ is screening-free and shifts only $\sim 15\%$. (*alpha_vac_axiom*, *foundational*, *registry.py:238*; *alpha_vac_derivation*, *open_negative*, *registry.py:274*.)

The adopted value is the framework's deepest open gap.

OVERCLAIM FLAG (Appendix A #21). Do not write " α is derived." The value is ADOPTED; only the conditional $a/c = 1/3$ math is proven, and its antecedent (the IR-carrier identification) is OPEN.

The Constitutive Hub

The four inputs assemble into one equation, the load-bearing hub every later sector is a regime of:

$$\tau_0 \dot{Z} + Z = Z_{\text{target}}[Z].$$

Q supplies the *form-class*: the CTP variation yields a causal, retarded, passive (Herglotz) relaxation structure — a response-to-the-past toward a target, with FDT fixing the noise — but only the form-class, not a single pole. F upgrades the form-class to the law: the MZ Markovian limit collapses the generic Herglotz form to the specific single pole, and the *value* τ_0 rides in here as an anchor. D supplies neither form nor law but the *structural reading* — it certifies that the one proper length $L_0 = c\tau_0$ is the controlled, non-anomalous breaking of scale-freedom and confines the nontrivial action to the tensor sector. α supplies the normalization, not the dynamics (the relaxation is α -independent). The hub is therefore **DERIVED-as-synthesis** [SPLIT]: the form is derived; the law-as-pillar, the τ_0 value, and α 's antecedent are taken in, and named. (*constitutive_equation, computed, registry.py:496; memory_kernel_form, computed, registry.py:516.*)

The Hierarchy of Layers

Read top to bottom, GRUT's ontology is a layered stack, and the chapters of this document walk it: **Q** → **F(t)** → **responsiveness** → **vacuum** → **physics** → **complexity** → **observers**. Q is the floor — scale-free, always present. Beneath the *vacuum*, but above the floor, sits the microscopic KMS bath **F(t)**: the substrate sector GRUT posits but does not yet specify, and the one place a genuine cold relic or a consciousness hook could attach. This layering matters for honesty: descending into F(t) to give it field content is GRUT extending into its *own* deeper layer — it breaks no premise and is not a successor theory. Only breaking Q or locality would constitute "a different theory, not GRUT."

The Tier Legend, in Action

Every emergence event below carries exactly one of five tags, mapping onto the machine-checked registry tiers as set out in the boundary legend: **DERIVED** (made from GRUT's own structure, zero/low free parameters); **HOSTED** (received as a boundary condition GRUT does not generate); **FORBIDDEN-BY-THEOREM** (ruled out from GRUT's own axioms); **OPEN** (no mechanism yet, but attackable); **CONJECTURAL** (a structural hook only, far from derivation). Many events are **split-tier seams** tagged [SPLIT] — mechanism DERIVED but input anchored, or form DERIVED while antecedent OPEN — and the split is the honest content; it must never be collapsed to the higher tier. The whole apparatus is itself auditable: a 121-claim registry (verified composition: 48 computed, 28 open_negative, 20 anchored, 11 conjectural, 10 meta, 4 foundational), each claim carrying a tier, code references, tests, and a single falsifier, with over 3,000 gating tests; a claim advances a tier only when a test certifies it, and a failed result is demoted and logged, never quietly dropped. That meta-layer — the thing that makes the tiering auditable rather than asserted — is itself **DERIVED**.

And the seams are already visible at this rung, before the story has begun. **Dark matter is HOSTED** — received, not derived, by the locality/no-halo and single-mode results — and it is a *resident of GRUT's own substrate sector F(t)*, internal to the layered ontology, not external and not a successor theory. **A propagating dark relic and the hierarchy magnitude are FORBIDDEN-by-theorem**, and (as Part IV shows) for the *same* reason. **The α -antecedent is OPEN, τ_0 is anchored, and consciousness is at most a CONJECTURAL hook**. None of it is hidden. That is the legend. Now the universe runs forward.

Chapter 3 — Genesis of Responsiveness

Run the clock backward as far as it will go, and you do not arrive at nothing. You arrive at **Q without F** — a medium that already responds to realized differences, with the causal arrow already pointing from past to future, but with *no memory, no characteristic length, and therefore no time in the sense of a clock the universe sets for itself*. This is the deepest positive result in the framework and the keystone of the layered ontology: **across every backward crossing, Q is the one invariant that survives the loss of memory**. It contains no τ_0 , no τ_{micro} , no α ; it is the floor everything else rides on. The arrow of time does not emerge *with* responsiveness — it predates it. — **DERIVED**. (GRUT_GENESIS.md §3; ctp_action_structure, computed; the MZ and pole-spectrum verify harnesses pass.)

Spec — Q survives the loss of memory (the keystone). Tier: DERIVED. On the closed-time-path contour, the influence action vanishes on the diagonal, $S_{\text{IF}}[\varphi_+ = \varphi_-] = 0$, and physical response is its variation in the difference of the two history branches. Q carries no τ_0 , no τ_{micro} , no α — it is scale-free. How: run the clock back across every memory-losing crossover and the single invariant that survives is this in-in causal arrow (response-to-past, never-to-future). The arrow predates memory; everything above is Q wearing more structure. (ctp_action_structure, computed, registry.py:451; highest-fanout node, 93 dependencies.)

The Pre-Responsive Substrate

What exists in this Q-only phase, besides Q itself, is the microscopic KMS bath **F(t)** — a white-noise bath with passive response and no slow collective variable to carry memory. Above a critical temperature T_c the prospective slow shear variable z is *not slow relative to its own bath*: $k_{\text{BT}} \gtrsim \hbar/\tau_{\text{micro}}$, so there is no slow/fast separation for a Mori–Zwanzig projection to act on, and no memory. The medium is fluctuating but memoryless. The arrow is present — Q's retarded structure acts directly on F(t), needing no τ_0 and no slow z — but the universe *necessarily forgets*, because the only content is the un-projected bath. (The microscopic scale itself, $\tau_{\text{micro}} \equiv \hbar/k_B T_c$, is registry-tiered **conjectural** (tau_micro_thermal_scale, registry.py:1442) and is reported here as hosted-with-forbidden-relation-to- τ_0 — the load-bearing two-anchor seam, returned to in Part IV.)

The Emergence of Memory

Now let the universe cool. As the temperature falls through T_c , the slow/fast separation opens, and one relaxation channel — the single transverse-traceless shear mode locked to the conserved stress-energy ($\partial_\mu T^{\mu\nu} = 0$ being the diffeomorphism Ward identity, part of Q) — fails to equilibrate. **Memory condenses as the un-rejected remainder**: the MZ projection of that surviving slow channel yields the first-order, history-retaining response $\chi = 1/(1 - i\omega\tau_0)$ as its Markovian limit. The *existence* and the *slow, gapless character* of this channel are forced, without tuning, by conservation. — **DERIVED** [SPLIT]: existence and character forced; the finite-temperature MZ calculation that would rigorously prove memory turns *off* again above T_c is asserted via the $\tau_K < \tau_0/4$ threshold, not yet computed — the forward story uses the temperature law $f(T)$ as a proxy for that uncomputed dynamics. (memory_kernel_form, computed; GRUT_GENESIS.md §8.1, the finite-T MZ collapse named as "the decisive uncomputed dynamics.")

How fast does memory switch on? Not by an imported sigmoid with a free width, but by GRUT's own noise ratio. Splitting Q's FDT/KMS relation into its zero-point and thermal parts gives the coherent fraction of the bath

fluctuation, and that fraction is the order parameter:

$$f(T) = \tanh\left(\frac{T_c}{2T}\right),$$

and composing with radiation-era cooling $T(t) = T_c (t_c/t)^{1/2}$ gives the explicit forward switch-on profile $\bar{z}(t) = \tanh[1/2 (t/t_c)^{1/2}]$. This is the headline new v4 result the older backward catalog never wrote. It does *not* equal a clean $1/2$ at T_c : $f(T_c) = \tanh(1/2) = 0.46212$, with a power-law $1/T$ high-temperature tail rather than an exponential one. — **DERIVED** [SPLIT]: the shape and limits follow from Q's FDT/KMS structure given *one physical identification* (memory-activation = the bath's zero-point fraction at the single τ_{micro} scale). It trades the old sigmoid's free width for a motivated identification — an improvement, *not* a parameter-free theorem; the complement $1 - \tanh$ or a τ -ratio $T_c/2\pi T$ are a-priori alternatives. One caveat kept sharp: it is the memory-*amplitude* order parameter $f(T)$ that is derived this way; the elastic *rigidity* onset shape (the G_0 of v4's solid candidate) is, per GRUT_V4_ELASTIC_VACUUM.md §7, still an ad-hoc sigmoid pending the finite-T MZ calculation — so "{F, D-breaking, G_0 }" share one order parameter" is an aspiration on the rigidity leg, not a derivation.

And here the theory corrects itself — exactly the discipline the spine is built to honor. The older catalog called this onset a "phase transition." The forward analysis demotes it. The pole *amplitude* $f(T)$ is C^∞ -analytic straight through T_c — no critical onset, no diverging susceptibility, no latent heat — so by Landau's analyticity criterion it is a **smooth crossover, not a second-order transition**. There is a sharp bifurcation in the underlying physics — the pole-*existence* boundary at $\tau_K = \tau_0/4$ — but it sits roughly 34 orders of magnitude away ($(\tau_K(T_c)/(\tau_0/4)) \approx 7 \times 10^{-35}$), structurally pre-satisfied and never reached in cosmic history. Emergence is a phase transition *only* in the spontaneous-scale-symmetry-breaking sense, $L_0: 0 \rightarrow \text{finite}$ — executed as an analytic amplitude crossover, decidedly not as a critical quench. — **DERIVED**. (GRUT_GENESIS.md §8.2.)

The Emergence of the Responsive Vacuum

The same condensation that gives z its amplitude gives it a finite memory time τ_0 , and therefore exactly one proper length, $L_0 = c\tau_0 \approx \mathbf{12.85 \text{ Mpc}}$. "Capable of memory" and "scale-free" are now mutually exclusive: the appearance of L_0 breaks the boundary charge of the dilatation redundancy D at $O((L_0 k)^2)$, non-anomalously ($4 \nmid \{2, 6, 10, 14\}$), with a measure Jacobian identically one (so α does not enter). The decisive falsifier — that the Keldysh doubling might secretly spoil the redundancy — is discharged: T_λ acts diagonally on the $\pm$ branches, $|J|^2 = 1$, and the influence functional collapses to two decoupled diffeomorphism-invariant GR copies as $L_0 \rightarrow 0$. The emergence of responsiveness is this breaking of D by the crystallization of one proper length: spontaneous scale-symmetry breaking in the vacuum's response sector. — **DERIVED** [SPLIT]: dynamics and measure **DERIVED**; the initial state is theorem-modulo GAP-1 (the in-in state must be spatially homogeneous and scale-free — the standard adiabatic / $n_s = 1$ / cosmological-principle condition), whose floor is the field-wide cosmological measure problem, shared by all of cosmology, not specific to GRUT. One notch short of "forced from Q." (GRUT_GENESIS.md §§6–7; GRUT_V3_ORGANIZING_STRUCTURE.md *Bridge D*.)

Why Exactly One Relaxation Channel Survives

The deepest question of the genesis program: how does a Q-only medium develop a long-lived relaxation mode — why does *one* channel fail to equilibrate, with $\tau_0 \gg \tau_{\text{micro}}$ — without inserting the hierarchy by hand? The answer divides cleanly into what is forced and what is not.

Forced (no number tuned). The *existence* of the slow channel: the TT/shear metric memory is the response to the *conserved* stress-energy $T_{\mu\nu}$, where $\partial_\mu T^{\mu\nu} = 0$ is the diffeomorphism Ward identity — enforced by the same transverse projector P^{TT} that defines the response. Conservation is part of Q , so the slow channel's existence is not assumed but forced. Its *slow, gapless character*: locked to a conserved current, the mode relaxes only by transport, hence is hydrodynamically gapless ($\omega \rightarrow 0$ as $k \rightarrow 0$). And the hierarchy must be *large*: the MZ existence condition makes $\tau_0 \gg \tau_{\text{micro}}$ the validity condition of the slow/fast projection itself — the one channel's failure-to-equilibrate is the retained history. So GRUT forces *why* one channel survives, and forces that the gap is large — the qualitative half of the central question, without tuning.

Not forced — the honest miss. The *size* (the 34 orders; $c = \ln(\tau_0/\tau_{\text{micro}}) = 78.233$) is not forced. It relocates verbatim into the IR length L_0 and stays anchored, expressible as a difference of two independently-anchored Planck-referenced logs: $\ln(\tau_0/t_P) - \ln(\tau_{\text{micro}}/t_P) = 134.447 - 56.214 = 78.233$ (the two logs are differenced before rounding). Not even the exponential (dimensional-transmutation) *form* is instantiated: GRUT has no β -function carrying τ_0 ; its only RG statement is that the τ_0 IR scale is *RG-protected*, the opposite of an asymptotic-freedom flow. The tempting $8\pi^2 = 78.957$ (+0.92%) is rejected by GRUT's own near-miss standard — it is the universal one-instanton action, not a GRUT-distinctive constant, and GRUT has no derived tunneling linking the τ_{micro} and τ_0 sectors. — **FORBIDDEN-BY-THEOREM** (magnitude) + **OPEN** (antecedent).

Spec — The two-anchor hierarchy. Tier: existence FORBIDDEN-to-vary / magnitude anchored. The 34-order gap $c = \ln(\tau_0/\tau_{\text{micro}}) = 78.233$ is exactly the difference of two independently-anchored Planck-referenced logs: $\ln(\tau_0/t_P) - \ln(\tau_{\text{micro}}/t_P) = 134.447 - 56.214 = 78.233$ (differenced before rounding), **not** a third number. How: the MZ slow/fast split forces $\tau_0 \gg \tau_{\text{micro}}$ (existence), but no β -function carries τ_0 , so the size is taken in, not made — everything else hierarchy-flavored is a disguise of these two anchors. (GRUT_HIERARCHY_LEDGER.md §4.)

The honest miss is the size, not the existence.

OVERCLAIM FLAG (Appendix A #2, #15). τ_0 's magnitude is the single fact that sinks the spine if it is ever inflated — existence is forced, size is not. And $f(T)$ trades a free width for a motivated identification, not a parameter-free theorem.

What Is Carried Through, Not Made

Two things ride *through* genesis rather than being made by it. The first is τ_0 's numerical value — the hierarchy just discussed, carried as an unexplained anchor. The second is the **$\alpha = 1/3$ spine** ($S = 108\pi = 339.292$, $R = \sqrt{4/3} = 1.15470$, with the dark-energy fraction Ω_Λ tied to $(2 - R)^2$): a dimensionless anomaly-sector relic, orthogonal to the dilatation sector, frozen *before* responsiveness and inherited by the GRUT phase rather than generated by it. — **DERIVED** [SPLIT]: everything downstream of α is derived; the derivation of α is OPEN. (GRUT_GENESIS.md §3; *alpha_vac_axiom, foundational; screening_108pi, computed.*)

Of everything genesis leaves behind, exactly one relic is already observed: **dark energy is the Herglotz spectral weight of the single pole, tied to τ_0** . It switches on adiabatically as $\omega\tau_0$ drops through 1 over the whole subsequent expansion, with no localizable transition epoch — the only downstream fingerprint of the genesis itself. The identification is the load-bearing claim; the precise number carries the anchored caveat (the raw tree relation gives $(2 - R)^2 = 0.71453$; the full-Friedmann anchored value is 0.6886 — two different numbers, never conflated). — **DERIVED** (the emergence) / *anchored* (the value). (GRUT_GENESIS.md §3; *omega_lambda_prediction, anchored.*)

Spec — Dark energy as terminal velocity. Tier: mechanism DERIVED, number anchored.
*The expansion drive (a negative-energy conformal mode) balances the vacuum's memory friction at a constant rate — a terminal velocity $H_{\text{inf}} = (2-R)/(S\tau_0)$. Anchored full-Friedmann value: $\Omega_\Lambda = 0.6886$ (Planck 0.6889; 0.04% match). How: dark energy is the Herglotz spectral weight of the single memory pole tied to τ_0 . **Discipline:** the tree closed form $(2-R)^2 = 0.71453$ ($R = \sqrt{4/3} = 1.15470$) is a cosmic-baseline approximation, **not** the anchored 0.6886 — never conflate them. (omega_lambda_prediction, anchored, registry.py:1346.)*

The Blind Census and the Honest Negative

Then the test that defines the boundary. Run the relic inventory forward *without* biasing it toward producing dark matter, and ask whether viable cold dark matter falls out. It does not. The single-mode structure forbids a second stable *relaxational* pole — itself a channel-counting postulate, not a consequence of Q alone, the seam laid bare in Part IV. The one condensation-tied relic — an elastic shear phonon at $k_{BT_c} = 4.71$ keV — is *warm*, travels at $c_s = c$, and overcloses the universe by a factor of several hundred to several thousand. GRUT does not predict *no* dark matter; it predicts the *wrong* dark matter, and reports that as a genuine negative. **Dark matter is HOSTED** — and it is a resident of GRUT's own substrate sector $F(t)$, the microscopic medium beneath the vacuum that current GRUT posits but does not yet specify. That descent into $F(t)$ breaks no premise; it is GRUT extending into its own deeper layer, not a successor theory. — **HOSTED.** (GRUT_GENESIS.md §4; locality_no_halo_theorem, computed.)

Spec — Dark matter is substrate-resident, not derived. Tier: HOSTED. *Run the relic inventory forward without biasing it toward dark matter and no viable cold DM falls out: the one forced relic is a warm 4.71 keV shear phonon ($c_s = c$) that overcloses by $\sim 418 \times - 5800 \times$. GRUT predicts the wrong DM and reports it as a genuine negative. The host is a coherent $\sim 10^{-22}$ eV scalar **resident in GRUT's own pre-responsive substrate $F(t)$** — internal to the $Q \rightarrow F(t) \rightarrow$ vacuum ontology, not external, not a successor. Honest cost: exactly 2 inserted dials (mass m , amplitude ϕ_i); 0 of 3 requirements sourced from the bath.*

The Qualitative Keystone

The genesis closes on one sentence, the keystone of the whole layered ontology:

Q is the one invariant that survives the loss of memory. *Responsiveness = (the always-present causal arrow Q) $\times$ (a single proper length L_0 breaking the dilatation redundancy D), unmasked as a stable single-pole memory the moment cooling through T_c made one gravitational mode slow relative to the vacuum bath. The switch is L_0 : $0 \rightarrow$ finite; the timer is the 34-order slow/fast gap opening as the universe cools.*

The universe became capable of memory not by acquiring a field or a force, but by spontaneously breaking a redundancy — and that breaking became *possible* the moment cooling opened a timescale hierarchy. The arrow (Q) was always there; only the *retention* (L_0) switched on. With memory in hand, the universe now has a clock, a vacuum, and — for the first time — a history. Part II is what that history builds.

PART II — EMERGENCE OF PHYSICS

$Q \rightarrow F(t) \rightarrow \text{Responsiveness} \rightarrow \textbf{Vacuum} \rightarrow \textbf{Physics} \rightarrow \text{Complexity} \rightarrow \text{Observation} \rightarrow (\text{the universe discovers responsiveness}).$

Part I left the universe with a responsive vacuum: a medium that has crystallized a single memory time τ_0 out of a pre-responsive substrate, breaking scale-freedom by the one proper length $L_0 = c\tau_0$. Nothing yet *does* physics. This Part is the next stretch of the ladder — the rungs **Vacuum** $\rightarrow$ **Physics** — in which the responsive vacuum, once it exists, turns out to behave like a field theory, and the known sectors of nature emerge not as separate postulates but as *regimes of one constitutive law*.

A word on ordering, because it is the whole point of this restructuring. This is a **developmental** sequence, not a physics-textbook one. The vacuum is named first (Ch 4) because everything downstream is a regime of it. Quantum mechanics (Ch 5) comes next because it is the *shortest-memory* face — the $\tau \rightarrow 0$ limit — the regime closest to the bare constitutive step. Gravity (Ch 6) is the elastic/curvature response of the same medium. Cosmology (Ch 7) is that medium read at the largest scale, where its single memory pole leaves the one relic we actually observe. The universe builds the apparatus, then runs it from fast to slow, from local to cosmic.

Throughout, the tier legend governs every cell. The two facts that must never be inflated in this Part are flagged where they live: the **Born weights** (OPEN, in Ch 5) and the **Ω_Λ number** (anchored 0.6886, never the tree-level 0.71453, in Ch 7).

Chapter 4 — The Responsive Vacuum

Before the universe can have quantum behavior, gravity, or a cosmic history, it must have the *object* all three are regimes of: the responsive vacuum and its one constitutive law. This chapter establishes that object and its honest tier. The headline tier is a [SPLIT] — the *form* of the law is DERIVED; the law-as-pillar and the scale τ_0 are HOSTED, taken in rather than made.

The memory kernel. The responsive vacuum relaxes on a single time $\tau_0 \approx 41.9$ Myr. The same physical fact has three equivalent faces, related by Fourier transform with no new content added between them: the memory kernel $\mathbf{K}(\mathbf{t}) = (\mathbf{1}/\tau_0) \mathbf{e}^{\{-\mathbf{t}/\tau_0\}} \Theta(\mathbf{t})$, the susceptibility $\chi(\omega) = \alpha/(1 - i\omega\tau_0)$ with one pole at $\omega = -i/\tau_0$ in the lower half-plane, and the constitutive law $\tau_0 \dot{\mathbf{z}} + \mathbf{z} = \mathbf{z_target}$ (the FT identity is verified in sympy, edge E5 — flagged as a *definitional identity, not a derivation step*). The single pole sits in the lower half-plane, making the kernel causal, Kramers–Kronig compatible, and recovering GR as $\omega \rightarrow \infty$. — **DERIVED** [SPLIT]: the form is the Mori–Zwanzig Markovian limit (derived); the value τ_0 rides in as an anchor (`memory_kernel_form`, `registry.py:516`, computed; `first_order_from_mori_zwanzig`, `registry.py:5422`, computed).

Spec — The constitutive law (the load-bearing hub). Tier: **DERIVED-as-synthesis** [SPLIT].

Every sector is a regime of one law: $\tau_0 \dot{\mathbf{z}} + \mathbf{z} = \mathbf{z_target}[\mathbf{z}]$, equivalently the susceptibility $\chi(\omega) = \alpha/(1 - i\omega\tau_0)$ (one pole at $\omega = -i/\tau_0$, lower half-plane) and the kernel $K(t) = (1/\tau_0)e^{\{-t/\tau_0\}}\Theta(t)$ — three faces of one object. How: Q supplies the causal/passive form-class; the Mori–Zwanzig Markovian limit collapses it to the single pole; the value τ_0 rides in as an anchor; α sets only the DC amplitude $\chi(0) = \alpha$. **Split:** the form is derived, the law-as-pillar and τ_0 are inputs — single-pole-ness is not a theorem of Q alone (`registry.py:5287`). (`constitutive_equation`, computed, `registry.py:496`.)

The constitutive law as the load-bearing hub. This is the single most important structural claim of the whole framework: **every sector is a regime of the one law $\tau_0 \dot{\mathbf{z}} + \mathbf{z} = \mathbf{z_target}[\mathbf{z}]$** . It is worth being exact about what supplies what, because the premise-set is the honesty:

- **Q supplies the form-class.** The CTP variation $\delta S_{\text{CTP}}/\delta \mathbf{z}_a|_{\{\mathbf{z}_a=0\}} = \mathbf{F}[\mathbf{z}_r]$ yields a *causal, retarded, passive* (Herglotz) relaxation structure — a response-to-the-past toward a target, with FDT fixing the noise. This is genuine and zero-input, but it is only the form-class; it does **not** by itself pick out a single pole (edge E1 SOLID).
- **F upgrades the form-class to the law.** The Mori–Zwanzig Markovian limit of one slow dissipative variable collapses the generic Herglotz form to the specific single pole, exact to $O(\tau_K/\tau_0)$. An off-axis "dark-capable" pole would require $\tau_K > \tau_0/4$, which GRUT's anchored hierarchy ($\tau_{\text{micro}}/\tau_0 \sim 10^{-34}$) forbids. The *value* τ_0 enters here as an anchor (edge E2 COMPUTED).
- **D supplies the structural reading, not the law.** The broken dilatation redundancy certifies that the hub's one proper length $L_0 = c\tau_0$ is the controlled, non-anomalous breaking of scale-freedom, and confines the law's nontrivial action to the **tensor** sector. That confinement is what later makes $\mu_{\text{linear}} = 1$ clean (edge E3 DOTTED).
- **α supplies the normalization, not the dynamics.** α sets the DC amplitude $\chi(0) = \alpha$ and fixes $S = 12\pi/\alpha^2$. The τ_0 relaxation is α -independent.

So the hub is **DERIVED-as-synthesis** [SPLIT]: the *form* is derived; the law-as-pillar, the τ_0 value, and α 's antecedent are taken in — named, not hidden (`constitutive_equation`, `registry.py:496`, computed). **FLAG:** the single-pole form is the MZ Markovian limit (derived); the law-as-pillar plus τ_0 are inputs. Single-pole-ness is *not* a theorem of Q alone — the passive $\square$ positive-Debye converse is false (`registry.py:5287`). Do not lift this to "F

derived from Q."

Locality — the rule that later forbids halos. The kernel is *local in the matter fields*: analytic and pole-free in k^2 near $k = 0$, as a causal local memory kernel must be. This is a structural property carried by the vacuum from the start, and it is the seed of a no-go theorem two chapters down — an extended $\rho \propto 1/r^2$ halo would require a $1/k^2$ pole (an inverse-Laplacian $1/\nabla^2$) that a local kernel cannot contain. The locality of the responsive vacuum is named here as a *property of the object*; its consequences are tiered in Ch 6 and Part IV. — **DERIVED**.

Ward identities. The vacuum's response is locked to a conserved stress-energy: $\partial_\mu T^{\mu\nu} = 0$ is the diffeomorphism Ward identity, and it is *part of Q* — not an extra assumption. This is what guarantees that the single relaxation channel which survives genesis is the transverse-traceless shear mode tied to conservation: the existence and slow, gapless character of that channel are *forced, without tuning, by conservation*. The Ward identity is also the seam that later (Ch 7) confines the modification to the tensor sector. — **DERIVED**.

The thermal activation of the kernel. The memory kernel is not always on. Above a critical temperature $T_c \approx 54.7$ MK the prospective slow shear variable is not slow relative to its own bath — there is no slow/fast separation, hence no memory. As the universe cools through T_c (≈ 16.5 hours after the Big Bang) the single-pole kernel activates and gravity becomes bandwidth-limited, with the refractive index running to $n_g \rightarrow R = \sqrt{4/3}$ at DC. The transition's *existence* and its $n_g \rightarrow R$ consequence are derived; its *value* T_c is fixed empirically by the standard cosmological-chronology pin. — **DERIVED** [SPLIT]: transition derived, value anchored (`t_c_thermal_transition`, `registry.py:1397`, computed). The activation profile is $f(T) = \tanh(T_c/2T)$, with $f(T_c) = \tanh(1/2) = 0.46212$, $f \rightarrow 1$ deep below T_c , and a $1/T$ power-law tail above — its shape forced by Q's FDT/KMS structure given one physical identification (activation = the bath's zero-point fraction). — **DERIVED** [SPLIT]: motivated identification, not a parameter-free theorem.

Spec — Memory switch-on profile. Tier: DERIVED [SPLIT]. *The memory-amplitude order parameter is $f(T) = \tanh(T_c/2T)$, with $f(T_c) = \tanh(1/2) = 0.46212$ (notably not $1/2$), $f \rightarrow 1$ deep below T_c , and a $1/T$ power-law (not exponential) high- T tail. Composed with radiation-era cooling it gives the forward switch-on $\bar{z}(t) = \tanh[1/2(t/t_c)^{1/2}]$. How: splitting Q's FDT/KMS relation into zero-point and thermal parts makes the coherent bath fraction itself the order parameter. **Split:** this trades the old sigmoid's free width for a motivated identification — an improvement, not a parameter-free theorem.*

Why responsiveness looks like a field theory. The CTP variation does not merely permit a field-theoretic description — it *produces* one. The form-class that drops out is causal, retarded, and passive (Herglotz); the noise is fixed by FDT with $N \geq 0$; the response is tied to a conserved current by the Ward identity. Those are precisely the structural features of a unitary field theory's two-point function. The responsive vacuum looks like a field theory because the in-in formalism is the field-theoretic generating structure, read for response rather than amplitudes. This is the object on which all of Part II's physics rides. **FLAG:** the law-as-pillar and τ_0 are taken in — edge E2 is COMPUTED, not SOLID; do not present the vacuum as "derived from Q" without the split.

Chapter 5 — Emergence of Quantum Reality

Run the constitutive law to its shortest-memory limit and quantum mechanics falls out. This chapter is almost entirely **DERIVED-as-recovery** — you get known quantum behavior back in a limit, you do not add a new postulate — with one razor-sharp seam where the boundary shows through: the **Born weights are OPEN**.

Quantum mechanics as the $\tau \rightarrow 0$ face of the vacuum. Write the constitutive law $\tau \dot{z} + z = z_{\text{target}}$ with z_{target} built from the Schrödinger residual as a Newton–Raphson step. In the $\tau \rightarrow 0$ (zero-memory) limit the equation becomes algebraic, $z = z_{\text{target}}$ — the instantaneous response — and one constitutive step is the first-order Euler–Schrödinger update exactly, with norm preserved and $\langle \sigma_x(t) \rangle = \cos \omega t$ on a precessing qubit. Quantum mechanics is the zero-memory face of the responsive vacuum. — **DERIVED** (as recovery, not from nothing): the recovery is exact *as a limit*, verified at first order on a single qubit; promotion to many-body Hamiltonians is the unbuilt part (qm_recovery, registry.py:753, computed; edge E10 SOLID). **FLAG:** "GRUT derives QM" means *recovers QM in a limit, at first order on one qubit* — not a from-scratch construction and not yet many-body.

The arrow of time — recovered, and more primitive than memory. Constitutive evolution carries an entropy production $\dot{S} = (1/\tau_0)(z - z_{\text{target}})^2$, non-negative for any state, zero only at the fixed point, cumulatively monotone — verified numerically strictly positive over 10^5 random states and exactly zero at $z = z_{\text{target}}$. The Second Law follows from Q's retarded kernel rather than being postulated alongside it (arrow_of_time_from_entropy, registry.py:1734, computed). And the directionality is even more primitive than memory: the in-in arrow (response-to-past, $S_{\text{IF}}[\varphi_+ = \varphi_-] = 0$) is a theorem of the formalism containing no τ_0 , so it *predates* responsiveness — the Part I keystone reappearing as the floor of this chapter. — **DERIVED**. This $\dot{S} \geq 0$ result is also the only hook Part III's "Emergence of Life" chapter is allowed to use; it is a *precondition* for dissipative organization, nothing more.

***Spec — The arrow of time from the kernel. Tier: DERIVED.** Constitutive evolution carries entropy production $\dot{S} = (1/\tau_0)(z - z_{\text{target}})^2$ — non-negative for any state, zero only at the fixed point, cumulatively monotone (verified strictly positive over 10^5 random states, exactly zero at $z = z_{\text{target}}$). How: the Second Law follows from Q's retarded kernel rather than being postulated. The directionality is even more primitive than memory — the in-in arrow $S_{\text{IF}}[\varphi_+ = \varphi_-] = 0$ carries no τ_0 and predates responsiveness. (arrow_of_time_from_entropy, computed, registry.py:1734.)*

Measurement — dissolved into physics, almost entirely. When a deep-crystal apparatus ($\Lambda_{\text{grav}} \cdot \tau_0 \gg 1$) couples to a quantum object ($\Lambda_{\text{grav}} \cdot \tau_0 \lesssim 1$), the joint dynamics is apparatus-dominated and the faster crystallizer drags the slower across the refractive→classical threshold. Collapse is *physical contact* through the low-bandwidth memory channel — no observer postulate, the same scaling law applying to measurer and measured (verified on a 1 g / atom example, the apparatus crystallizing $\sim 10^{32}$ times faster). Wigner's friend dissolves as conditional-state consistency (measurement_resolution, registry.py:1792, computed; wigner_friend_dissolution, registry.py:1947). — **DERIVED** [SPLIT]: this derives decoherence, diagonalization, and the pointer basis — but **not** the probability weights.

The Born-rule seam. This is the sharpest seam in the spine, and it must be stated without softening. The CTP / noise-kernel machinery produces off-diagonal decay and an asymptotically diagonal density matrix in the pointer basis, but it does **not** produce the specific weights

from the noise kernel. **The Born rule is OPEN — and, honestly, HOSTED:** the weights are received from the postulated inner product, and no GRUT mechanism yet generates them

(born_rule_postulate_open_negative, registry.py:2120, open_negative; edge E10 scope flag). **FLAG**

(Appendix A #14): "GRUT derives QM probabilities" is the refuted-on-contact overclaim — do not make it.

Note in fairness that this is *not* a GRUT-specific weakness: Copenhagen, Many-Worlds, decoherent histories, and CSL all require extra structure at exactly this point.

***Spec — The Born-rule seam. Tier: OPEN (honestly HOSTED).** The CTP / noise-kernel machinery produces off-diagonal decay and an asymptotically diagonal density matrix in the pointer basis, but it does **not** produce the diagonal weights $|\langle\psi|\text{pointer}_i\rangle|^2$. How (the seam): those weights inherit from the postulated Hilbert-space inner product, not from the noise kernel — no GRUT mechanism yet generates them. Decoherence, diagonalization, and the pointer basis are DERIVED; the probabilities are received. (Not a GRUT-specific gap — Copenhagen, Many-Worlds, decoherent histories, and CSL all need extra structure here.)* (born_rule_postulate_open_negative, open_negative, registry.py:2120.)

What this chapter recovers, what it does not. Recovered as the zero-memory regime: unitary Schrödinger evolution (first order, one qubit), the arrow of time, decoherence, the pointer basis, the dissolution of the measurement and Wigner's-friend problems. Not recovered: the Born weights (OPEN/HOSTED), and many-body promotion (unbuilt). The seam is placed in the open, not papered over.

Chapter 6 — Emergence of Gravity

The same responsive vacuum, read in its *elastic and curvature* response, is gravity. This chapter has one clean derived win (dark energy as a terminal velocity — carried in full into Ch 7, where it belongs with cosmology), one near-term laboratory falsifier, one derived safety result, and one heavily-flagged premise about the *kind* of medium the vacuum is. The chapter's discipline: gravity *appears* geometric for derivable reasons, but the claim "gravity is the elasticity of a solid vacuum" is an **unpaid postulate**, not a derivation.

Why gravity appears geometric. The arrival of the single proper length $L_0 = c\tau_0$ is what makes gravity look like geometry. A scale-free theory has an adiabatic dilatation redundancy; the finite memory time gives the vacuum exactly one proper length, and a fixed proper length is logically incompatible with that redundancy. L_0 is therefore a *mass-like breaking of scale-freedom* — the one geometric scale the responsive vacuum carries. At high frequency the refractive index $n_g \rightarrow R$ in the refractive regime and $\rightarrow 1$ in the GR limit, so general relativity is recovered as $\omega \rightarrow \infty$. — **GR recovery DERIVED** (gr_recovery, registry.py:1071, computed).

The elastic response and G_0 — derived value, unpaid premise. v_3 used only the *viscous* half of the viscoelastic vacuum (the single-pole Maxwell-fluid memory). v_4 activates the *elastic* half: if the vacuum is a viscoelastic *solid* with static shear rigidity $G_0 > 0$, the Debye / Kleinert world-crystal identity forces $G_0 = \hbar / (c^3 \tau_{\text{micro}}^4) \approx 1.03 \times 10^{16} \text{ Pa}$ from $\{\hbar, c, \tau_{\text{micro}}\}$ with no dial — a real, mechanism-backed number. But the tier is demoted on a structural count: **the solid is not forced**. v_3 's memory is a Maxwell *fluid* (static TT shear $\rightarrow 0$), so the solid is one of at least four degenerate medium classes and $G_0 > 0$ is an **unpaid postulate** — the elastic-vacuum document is explicitly a candidate within the founding charter, not the settled thesis. —

DERIVED [SPLIT]: G_0 's *value* is derived from $\{\hbar, c, \tau_{\text{micro}}\}$; the viscoelastic-SOLID premise is one imported IOU. **FLAG (Appendix A #16):** do **not** call "gravity-from-defects" or "curvature-from-the-solid" derived — the SOLID premise ($G_0 > 0$) is an unpaid postulate. The elastic/curvature framing of gravity is conjectural-candidate, not established.

Spec — Vacuum rigidity G_0 . Tier: DERIVED [SPLIT]. *If the vacuum is a viscoelastic solid with static shear rigidity $G_0 > 0$, the Debye / Kleinert world-crystal identity forces $G_0 = \hbar / (c^3 \tau_{\text{micro}}^4) \approx 1.03 \times 10^{16} \text{ Pa}$ from $\{\hbar, c, \tau_{\text{micro}}\}$ with no dial. How (the split): the value is mechanism-backed and dial-free, but v_3 's memory is a Maxwell fluid (static TT shear $\rightarrow 0$), so the SOLID is one of ≥ 4 degenerate medium classes and $G_0 > 0$ is an unpaid postulate. Do not call "gravity-from-the-solid" derived.*

The decoherence plateau — the sharpest near-term falsifier. The same τ_0 that fixes the dark-energy fraction surfaces as a tabletop signature. For a gold microsphere of radius $1 \mu\text{m}$ (mass $\approx 80.8 \text{ pg}$), GRUT computes a decoherence frequency of **689 Hz** from $\{G, \hbar, c, \tau_0\}$ with zero free parameters, coherence destroyed in $\sim 1.5 \text{ ms}$ (decoherence_plateau, registry.py:1045, computed). Because the same τ_0 fixes Ω_Λ through the terminal-velocity relation, a tabletop measurement of where decoherence saturates pins both the memory scale and the dark-energy fraction at once. — **DERIVED.** The seam: 689 Hz is a *predicted test point*, not an anchor — it inherits τ_0 's anchoring, so measuring the plateau is a way of *measuring* τ_0 .

Spec — The decoherence-plateau falsifier. Tier: DERIVED. *For a gold microsphere of radius $1 \mu\text{m}$ (mass $m \approx 80.8 \text{ pg}$), GRUT computes a decoherence frequency of **689 Hz** from $\{G, \hbar, c, \tau_0\}$ with zero free parameters, coherence destroyed in $\sim 1.5 \text{ ms}$. How: the same τ_0 that fixes the dark-energy fraction surfaces as a tabletop signature, so a measurement of where decoherence saturates pins the memory scale and Ω_Λ at once. **Seam:** 689 Hz is a predicted test point, not an anchor — it inherits τ_0 's anchoring. (decoherence_plateau, computed, registry.py:1045.)*

Solar-System safety — a derived win, not an assumption. The same memory length must leave the Solar System untouched or the theory is dead on arrival. Because the refractive enhancement is bandwidth-suppressed, $\alpha_{\text{eff}}(\omega) = \alpha/(1 + (\omega\tau_0)^2)$ is far below measurement precision everywhere $\omega\tau_0 \gg 1$ — the entire Solar-System regime. GRUT passes **eight independent precision tests of GR** spanning > 10 orders of magnitude in frequency (Saturn ranging, Mercury perihelion, lunar laser ranging, GPS, Hulse–Taylor pulsar, Cassini Shapiro delay, LIGO propagation, Earth ranging), with safety factors from 2.3×10^5 (Saturn) to 1.5×10^{35} (LIGO), median $\sim 10^{16}$, $n_g \rightarrow 1$ by the same bandwidth gate that switches off memory at high frequency (`solar_system_safety`, `registry.py:692`, computed). — **DERIVED.**

So gravity emerges as the elastic/curvature regime of the responsive vacuum: GR is recovered at high frequency (DERIVED), the dark-energy fraction and the 689 Hz plateau ride on the same τ_0 (DERIVED, the number anchored), the Solar System is safe (DERIVED) — and the deeper ontological claim that gravity is the elasticity of a solid vacuum is an unpaid postulate, flagged loud. The dark-sector no-gos that also live in this regime (no-halo, no propagating relic, hierarchy magnitude) are placed in Part IV, where the 5-point format keeps them honest.

Chapter 7 — Emergence of the Cosmic Universe

Read the responsive vacuum at the largest scale, over the whole subsequent expansion, and you get cosmology. This is physics at the cosmic rung, riding on the vacuum. The chapter contains the framework's **one clean cosmological derivation** (linear cosmology = Λ CDM exactly), its **one observed relic** (dark energy as the memory pole's spectral weight, the *number* anchored), and a clear ledger of what is recovered versus what is hosted or open. The discipline that governs the whole chapter: **never conflate the tree-level $(2 - R)^2 = 0.71453$ with the anchored 0.6886**.

Expansion and dark energy — the mechanism DERIVED, the number anchored. Empty spacetime has a negative-energy conformal (Gibbons–Hawking) mode that drives runaway expansion; the vacuum's finite-memory kernel has a dissipative part that opposes rapid change. The two balance at a constant rate — exactly a **terminal velocity**: $H_{\text{inf}} = (2 - R)/(S \cdot \tau_0)$, the drive $(2 - R)$ being conformal-mode physics and the friction $1/(S \cdot \tau_0)$ the screened constitutive relaxation. The cosmological constant is then not a tuned vacuum energy but a balance, and $\Omega_{\Lambda} = (H_{\text{inf}}/H_0)^2$ follows with zero free parameters in the conversion ($h_{\text{inf_decomposition}}$, registry.py:1325, computed; this is dark energy as the Herglotz spectral weight of the single pole). — **DERIVED [SPLIT]**: the mechanism and structure are derived; the *number* is anchored through τ_0 .

The two-number discipline (non-negotiable). The number must be stated carefully because this is the place the narrative most tempts an upgrade. The full-Friedmann treatment with anchored τ_0 gives $\Omega_{\Lambda} = 0.6886$ (Planck: 0.6889, a 0.04% match). The tree-level closed form $(2 - R)^2 = 0.71453$ ($R = \sqrt{4/3} = 1.15470$) uses a cosmic-baseline approximation in which τ_0 and S both cancel. **These are two different numbers.** Conflating them, or calling 0.6886 "derived from $(2 - R)^2$," is exactly the over-claim that sinks the spine (edge E7). — Ω_{Λ} *mechanism* DERIVED [SPLIT], Ω_{Λ} *number anchored* (omega_lambda_prediction, registry.py:1346, anchored). **FLAG (Appendix A #3)**: the anchored 0.6886 $\neq$ tree 0.71453; this is one of the two spine-sinkers — never inflate it.

The Hubble rate. From $\{\alpha, \tau_0\}$ the core yields $H_0 \approx 68.8 \text{ km/s/Mpc}$, sitting in the tension gap. — **anchored** ($h_0_{\text{prediction}}$, registry.py:1365, anchored).

*Spec — $\mu_{\text{linear}} = 1$, the cleanest theorem. Tier: DERIVED. In the linear scalar sector the tracefree transverse projector $P^{\wedge TT}$ annihilates the constitutive response ($\partial^{\wedge \mu} P^{\wedge TT} = 0$), so $\mu_{\text{linear}} = 1$ exactly: linear cosmology is Λ CDM ($\gamma = 1$, no gravitational slip). How: the result is α -free, τ_0 -free, purely structural — forced two independent ways and over-determined at $\sim 32\sigma$ by the low- ℓ CMB ISW data. Its mirror: any enhanced-growth branch ($\mu \neq 1$) is **FORBIDDEN-by-theorem** and audit-trail-only. (adiabatic_dilatation_redundancy_nogo, computed, registry.py:5227.)*

$\mu_{\text{linear}} = 1$ — the headline robust theorem. This is the cleanest derivation in the entire framework. In the linear scalar sector the tracefree transverse projector $P^{\wedge TT}$ annihilates the constitutive response ($\partial^{\wedge \mu} P^{\wedge TT} = 0$), so $\mu_{\text{linear}} = 1$ exactly: linear cosmology is Λ CDM, with $\gamma = 1$ and no gravitational slip. The result is α -free, τ_0 -free, and purely structural — drawing only on D's *derived* leg (tensor confinement), not its presupposed leg — and it is forced two independent ways and over-determined $\sim 32\sigma$ by the low- ℓ CMB ISW data (adiabatic_dilatation_redundancy_nogo, registry.py:5227, computed). — **DERIVED**. Its mirror image is a *forbidden* family: any enhanced-growth branch ($\mu \neq 1$, the old "refractive" modification) is forbidden by separate-universe / adiabatic-dilatation invariance and independently killed at $\sim 32\sigma$ by ISW. The whole v_2 family — enhanced growth, a linear dielectric dark sector with $\Omega_{\text{dm}} = \alpha$ — is now **audit-trail-only, recorded as**

dead. — FORBIDDEN-BY-THEOREM. FLAG (Appendix A #13): the old "enhanced growth / $\Omega_{\text{dm}} = \alpha$ " line is FORBIDDEN and audit-trail-only — do not resurrect it.

Structure formation — recovered background, hosted/standard fill. Because linear cosmology is Λ CDM, atoms, stars, galaxies, and large-scale structure form by standard astrophysics running on GRUT's recovered- Λ CDM background. GRUT's contribution is the *consistency* ($\mu_{\text{linear}} = 1$, BBN-era null, no DM refractive signature), not a derivation of stellar or galactic physics. — **HOSTED / standard** for the linear background's astrophysics; **OPEN** for nonlinear structure detail

(nonlinear_structure_formation_grut_consistency, registry.py:3392, open_negative). This rung is detailed in Part III, Ch 8; here it is named only as cosmological history riding on the vacuum.

The two primordial handles — opposite verdicts. Cosmology hands GRUT two quantitative inputs and they fare oppositely:

- **Baryogenesis (existence derived, magnitude hosted).** The CTP path is asymmetric (the canonical ratio $R \neq 1$), so a nonzero baryon-to-photon ratio is *forced* by that in-in asymmetry — the derived half. The magnitude, however, is hosted: the native structural factor $(2 - R_B)/S_B$ is off-spine and cosmetic (its numerator $(2 - R_B) = 0.982$ is $O(1)$; the full factor $\approx 1.7 \times 10^{-3}$ — and the zero of $(2 - R_B)$ sits at $R_B = 2$, *not* $R = 1$, so η_B does not vanish there), while $\eta_B \approx 6.57 \times 10^{-10}$ — about +7.7% from the Planck 6.1×10^{-10} — rides on hosted SM inputs (Jarlskog J_{CP} , electroweak nonequilibrium K_{neq}), with the match reverse-fit via a documented S_B re-choice (the canonical $R = \sqrt{4/3}$ worsens it to -7.3%). — **existence DERIVED / magnitude HOSTED**, re-tiered **ANCHORED** (baryogenesis_eta_b, registry.py:1696, anchored). **FLAG (Appendix A #4):** the 7.7% agreement must not be read as zero-parameter.
- **The scalar amplitude A_s (the honest negative).** GRUT has no inflationary epoch; all three computational paths for $A_s \approx 2.1 \times 10^{-9}$ fail badly (OU variance misses by $\sim 10^{10}$; naive inflationary substitution by $\sim 10^{109}$; the closest dimensional candidate $\alpha/S^3 \approx 8.5 \times 10^{-9}$ sits a factor of four away and is flagged a *clue*, not promoted). A_s is observation-anchored input. — **OPEN** (primordial_amplitude_zero_parameter_open_negative, registry.py:4464, open_negative). **FLAG (Appendix A #17):** A_s is not a prediction.
- **The spectral tilt n_s (weaker still).** GRUT's constitutive-dissipation form $n_s = 1 - 2(H\tau)^2/(1 + (H\tau)^2)$ can match Planck's 0.965 only by solving for $H\tau$ from the *observed* n_s — a fit, not a prediction, source-tagged [HYPOTHESIS]. The genuine forward content is the *running* $dn_s/d\ln k$ (CMB-S4-falsifiable), which still leans on a phenomenological input. — **CONJECTURAL. FLAG (Appendix A #18):** n_s is a fit, not a prediction.

Spec — Baryogenesis, existence vs. magnitude. Tier: existence DERIVED, magnitude HOSTED. The CTP path is asymmetric (canonical ratio $R \neq 1$), so a nonzero baryon asymmetry is forced by the in-in asymmetry — the derived half. The magnitude is hosted: $\eta_B \approx 6.57 \times 10^{-10}$ rides on SM inputs (Jarlskog $J_{\text{CP}} \times$ imported electroweak K_{neq}), and the +7.7% match to Planck (6.1×10^{-10}) was **reverse-fit** via a documented S_B re-choice — using the canonical $R = \sqrt{4/3}$ worsens the fit to -7.3%. The native $(2 - R_B)/S_B$ factor is off-spine and cosmetic (its numerator $(2 - R_B) = 0.982$ is $O(1)$; the full factor $\approx 1.7 \times 10^{-3}$). Re-tiered **ANCHORED**; not zero-parameter. (baryogenesis_eta_b, registry.py:1696.)

The cosmic ledger. What is *recovered*: linear cosmology = Λ CDM exactly (the headline robust theorem), GR at high frequency, dark energy's mechanism, H_0 in the tension gap, baryogenesis at the level of existence. What is *anchored*: the Ω_Λ number (0.6886), H_0 , T_c . What is *hosted/standard*: structure formation on the recovered- Λ CDM background. What remains *open*: nonlinear structure, A_s (honest negative), n_s (fit). The cosmos is the responsive vacuum read at the largest scale — one derived theorem, one observed relic, and a set of

seams placed exactly where a referee would push.

Transition to Part III. Part II has carried the ladder from the **Vacuum** to **Physics**: the responsive vacuum as the load-bearing hub (Ch 4), quantum reality as its zero-memory face (Ch 5), gravity as its elastic/curvature regime (Ch 6), and cosmology as the vacuum read at the largest scale (Ch 7). At each rung the recovery is real and the seams are placed in the open. The next stretch — **Physics** → **Complexity** → **Observation** — is where the framework is most tempted to overclaim and where the discipline is therefore strictest: Part III places atoms, life, mind, and intelligence as honestly-tiered *slots* on the same ladder, not as things GRUT derives today.

PART III — EMERGENCE OF COMPLEXITY

$Q \rightarrow F(t) \rightarrow \text{Responsiveness} \rightarrow \text{Vacuum} \rightarrow \textbf{Physics} \rightarrow \textbf{Complexity} \rightarrow \textbf{Observation} \rightarrow (\text{the universe discovers responsiveness}).$

Parts I–II carried the ladder from the floor (Q) up through the responsive vacuum and the recovered physical sectors. Part III is where the ladder leaves the territory GRUT *derives* and enters the territory a completed Theory of Everything must one day *place*. These four chapters — Structure, Life, Mind, Intelligence — are **placed slots on the emergence ladder, not GRUT derivations**. Each one names where the rung sits, what GRUT actually supplies at that rung (usually a background, a constraint, or a single structural hook), and — stated in its first paragraph — its honest tier today. The discipline of this Part is subtractive: its value is in saying plainly what is *not* derived. A ToE that quietly promoted "GRUT has a hook for X" into "GRUT explains X" would forfeit exactly the credibility the tier legend exists to protect. None of these chapters claim more than a slot and, at most, a hook.

Chapter 8 — Emergence of Structure

Tier of this chapter, stated up front: the linear background is Λ CDM by a DERIVED theorem ($\mu_{\text{linear}} = 1$); the astrophysics built on it is HOSTED / standard, GRUT-consistent. Atoms, stars, galaxies, heavy elements, and the first information-bearing systems are not derived here. They are standard astrophysics running on the cosmological background GRUT *recovers* — and GRUT's contribution at this rung is the consistency of that background and a set of null constraints, not an account of how a star forms.

This is the first rung where the division of labor becomes explicit: **GRUT hosts, standard physics fills.** The reason it can host cleanly is the cleanest theorem in the framework — $\mu_{\text{linear}} = 1$: at linear perturbation order, GRUT's cosmology is exactly Λ CDM, forced two independent ways and over-determined by data (`adiabatic_dilatation_redundancy_nogo`, `registry.py:5227`, `computed` $\rightarrow \mu_{\text{linear}} = 1$). This recovery is itself a DERIVED win; what rides on it is hosted. Because the linear background is Λ CDM and nothing else, every downstream structure-formation result of standard cosmology and astrophysics — the growth of density perturbations into the first halos, gas collapse, the first stars, stellar nucleosynthesis building the heavy elements, galaxy assembly, and eventually the chemically rich, information-bearing environments that the later rungs presuppose — proceeds unmodified. GRUT does not compute stellar structure or galactic dynamics, and it does not claim to. The background's numerical anchors ($\Omega_{\Lambda} = 0.6886$, H_0) are themselves *anchored*, not derived; only the structural $\mu_{\text{linear}} = 1$ theorem that licenses the hosting is DERIVED.

What GRUT *does* supply at this rung is twofold, and both pieces are constraints rather than derivations:

- **A protected early-universe history.** GRUT borrows standard Big-Bang nucleosynthesis and shows that its own constitutive correction *vanishes* there: BBN-era dynamics sit at $\omega_{\tau_0} \gg 1$, so $n_g \rightarrow 1$ exactly and the responsive vacuum leaves no refractive signature on primordial abundances. This is a **derived null** — GRUT does not compute BBN abundances; it borrows standard BBN and shows that its own correction vanishes there. The structure-forming era inherits an undisturbed initial chemistry.
- **No anomalous structure-scale signature.** The same locality that elsewhere forbids a dark-matter halo from the vacuum kernel means GRUT adds no new scale-dependent growth in the linear regime; linear cosmology remaining Λ CDM is precisely the statement that structure forms as in the standard picture.

The honest open edge here is the **nonlinear** regime. Whether μ_{GRUT} remains self-consistent once density contrasts reach $\delta \sim 10^2$ – 10^6 — halo formation, mode coupling, the regime where small systematic effects compound over N-body growth factors of $\sim 10^6$ — is an explicitly registered open question

(`nonlinear_structure_formation_grut_consistency`, `registry.py:3392`, **open_negative**). The linear structural proof does not extend itself into the nonlinear regime; answering it requires a modified-gravity N-body simulation that has not been run. So the honest tier for the *detailed* nonlinear structure ladder is **OPEN**, sitting on top of a linear background that is HOSTED-on-derived- Λ CDM.

Overclaim guard — DERIVED μ_{linear} , otherwise HOSTED/STANDARD. Do not present any astrophysical result — a stellar mass function, a galaxy luminosity relation, the existence of heavy elements — as GRUT-derived. The SM matter content that builds atoms is itself an input, not an output

(`sm_field_content_locked`, `registry.py:2552`; notes at `registry.py:2572`: "The SM field content is taken as input, not derived"). GRUT's role at the structure rung is exactly: it recovers the linear background as Λ CDM (a derived theorem), it protects the BBN-era chemistry with a derived null, and it adds no anomalous structure-scale signal. Standard physics does the rest.

Chapter 9 — Emergence of Life

Tier of this chapter, stated up front: OPEN slot, with one DERIVED hook. Life is not derived by GRUT, and no biological mechanism is claimed. This chapter places the rung on the ladder and names the single structural hook GRUT genuinely contributes — the derived arrow of time / entropy-production inequality — which is a *precondition* for dissipative self-organization, not an account of it.

On the emergence ladder, thermodynamic organization sits above stable chemical structure and below mind: it is the rung where matter starts to maintain low-entropy, memory-bearing, adaptively responsive configurations against the entropic tide. A completed ToE must eventually place this rung. GRUT, today, places it and supplies exactly one relevant hook, no more.

The hook — and only the hook — is GRUT's derived arrow. Constitutive evolution carries an entropy production $\dot{S} = (1/\tau_0)((z - z_{\text{target}})^2)$ that is non-negative for any state, zero only at the fixed point, and cumulatively monotone — the Second Law follows from Q's retarded kernel rather than being postulated alongside it (`arrow_of_time_from_entropy`, `registry.py:1734`, **computed**, verified over 10^5 random states and exactly zero at $z = z_{\text{target}}$). Crucially, the arrow is even more primitive than memory: the in-in directionality ($S_{\text{IF}}[\varphi_+ = \varphi_-] = 0$) contains no τ_0 and *predates* the emergence of responsiveness itself — it is the keystone of Genesis reappearing here as the floor of the complexity ladder. A directed, monotone entropy production is the thermodynamic precondition for any dissipative, self-organizing, memory-bearing system: without an arrow there is no gradient to feed on, no direction in which order can be locally maintained at the cost of global entropy export.

That is the entire GRUT content at this rung. The arrow is a precondition, not a mechanism. GRUT offers no model of metabolism, autocatalysis, replication, heredity, selection, or adaptation. It does not predict where, when, or whether life arises, and it does not derive any biological observable. Biology is an **OPEN slot**, placed on the ladder for map-completeness and connected to the framework by the single thread of the derived arrow.

Overclaim guard — MOSTLY OPEN. This chapter must read as "here is the slot and its one hook," never as "GRUT explains life." The arrow $\dot{S} \geq 0$ is the relevant hook and it is genuinely DERIVED; everything from there to a living system is unbuilt and unclaimed. The honest statement is one of placement: GRUT supplies the thermodynamic arrow that *permits* dissipative self-organization, and stops there.

Chapter 10 — Emergence of Mind

Tier of this chapter, stated up front: CONJECTURAL — a hook, not a theory of mind. GRUT does not explain observation, internal models, or consciousness. It contributes one *proven structural object* — the CTP in-in self-comparison — and the reading of that object as anything mind-like is a conjectural interpretation layered on top of it. The object is a theorem; the metaphor is flagged.

The mind rung is where the universe contains subsystems that build internal models and that *measure* — the rung the observer problem lives on. GRUT's structural foothold is genuine and worth stating precisely. A closed universe has no external observer, so it performs self-measurement by maintaining two copies of its own history — ϕ_+ forward, ϕ_- backward — and the physical dynamics is the *difference* between them: $S_IF[\phi_+ = \phi_-] = 0$, zero response when nothing has changed. The fixed point $z^* = z_target[z^*]$ is the state that generates its own boundary conditions (`closed_universe`, `registry.py:109`, **foundational**; `fixed_point_principle`, `registry.py:131`, **foundational**). **This object — the in-in self-comparison vanishing on the diagonal — is proven.** It is a genuine theorem of the formalism. Reading it as "the universe knowing itself" is a structural interpretation; the structure is proven, the reading is flagged (**CONJECTURAL [SPLIT]**).

The same closed/self-referential stance produces one further interpretive worked example — the "observer as the boxed system" inversion of Schrödinger's cat, where observation is synchronization through contact rather than creation through measurement (`schrödinger_in_box_inversion`, `registry.py:1854`, anchored as a *philosophical reformulation, not new physics*). And there is one numerical coincidence sometimes attached at this rung: a 40 Hz neural resonance that arises from two independent framework routes (Λ_grav at tubulin-dimer parameters $\rightarrow 39.9$ Hz; self-referential fixed-point network dynamics $\rightarrow 41.7$ Hz), sharing no fitted parameters. **The 40 Hz numbers are computed; the consciousness bridge is not.** The registry tiers the bridge **conjectural** and explicitly speculative (`neural_resonance_speculative`, `registry.py:2170`: "The 40 Hz number is computed from the framework; the consciousness mapping is interpretive only"; see also `observer_as_crystal`, `registry.py:1837`, **conjectural** — "Layer-2 interpretation").

Overclaim guard — CONJECTURAL, a hook not a theory of mind. "GRUT explains consciousness / the universe is conscious" is the cardinal overclaim of the entire framework (Appendix A #5). What the hook *is*: a proven self-reference structure plus a non-load-bearing numerical coincidence. What it is *not*: any account of experience, qualia, or awareness. What would be required to go further: a derivation connecting the fixed-point self-reference to a first-person observable, which GRUT does not have and does not claim. Nothing in the framework depends on this rung — if the consciousness reading is wrong, nothing else changes — and the v3 reader edition makes no consciousness claim at all. Honesty here is load-bearing for the whole ToE's credibility.

Chapter 11 — Emergence of Intelligence

Tier of this chapter, stated up front: CONJECTURAL / aspirational-philosophical, labeled as such. No derivation is claimed. This is the softest chapter in the spine and must read as a labeled closing reflection on the ladder, not a scientific claim.

This is the reflexive closure of the emergence ladder. Symbolic systems, science, and artificial intelligence are responsive systems that have begun to build models of responsiveness itself — the universe's crystallized medium constructing representations of the medium. Sitting atop the mind rung, intelligence is where the in-in self-reference (Ch 10) becomes explicit and externalized: the fixed-point structure $z^* = z_target[z^*]$, in which a system generates its own boundary conditions (`fixed_point_principle`, `registry.py:131`, foundational), is mirrored in subsystems that model, predict, and intervene on their own substrate. A theory of responsiveness, written by responsive systems, is the ladder folding back to study its own first rung.

That is the whole of the claim, and it is offered as reflection, not derivation. GRUT has no theorem about symbolic reasoning, scientific cognition, or artificial intelligence. There is no mechanism, no observable, no falsifier here — only the observation that the ladder, run to its top, produces systems capable of writing down the framework that describes the ladder. This reflexive closure is the narrative justification for the document's through-line ending — *the universe discovers responsiveness* — and it is exactly as strong as that: a coherent closing image, explicitly tiered as aspirational and philosophical.

Overclaim guard — ASPIRATIONAL/PHILOSOPHICAL. This is the softest rung on the ladder. It must be kept short and clearly labeled. Do not present the reflexive-closure image as a result; it carries no tier above CONJECTURAL and asserts no derivation. Its only role is to close the complexity ladder and hand the narrative to Part IV's open frontiers and Part V's synthesis: responsive systems have arrived at the point of studying responsiveness — which is precisely the program the rest of the document audits.

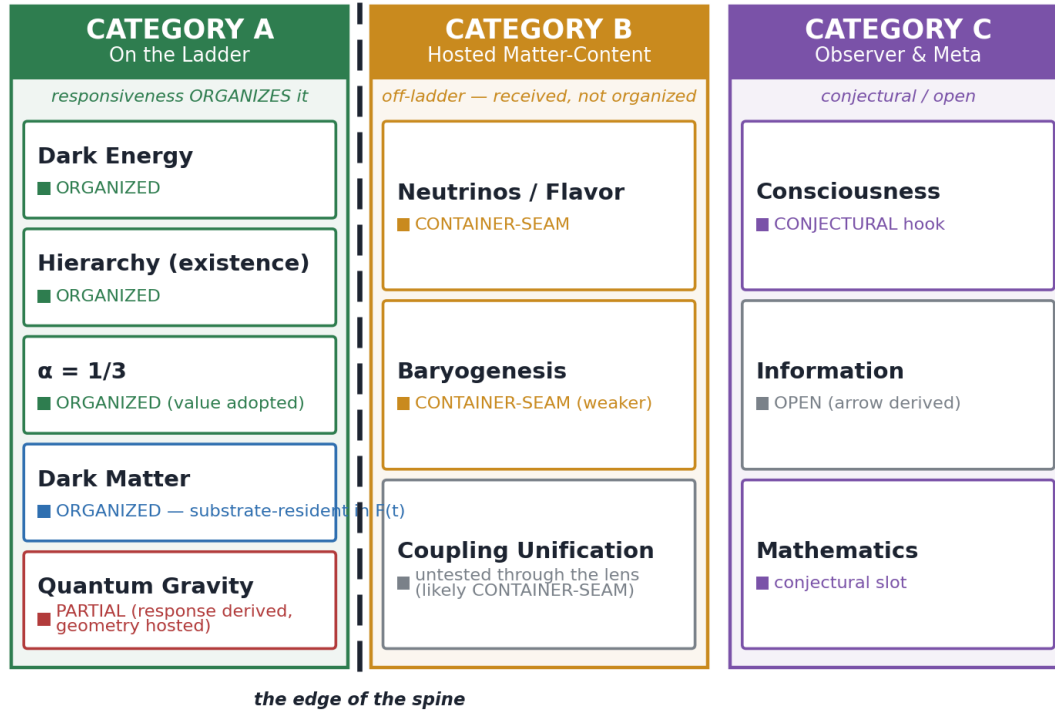
Part transition out: *The complexity ladder has carried the spine from recovered physics (Structure, HOSTED on derived- Λ CDM) through the thermodynamic precondition for life (Life, an OPEN slot on the derived arrow), to the structural hook for mind (Mind, a proven object under a CONJECTURAL reading) and the reflexive closure of intelligence (aspirational). At every rung the discipline held: a placed slot and an honest tier, never a borrowed derivation. **Part IV now takes the open rungs — including the ones touched here — and subjects each to the same 5-point frontier audit that prevents any hook from inflating into a claim.***

PART IV — THE OPEN ToE FRONTIER

$Q \rightarrow F(t) \rightarrow \text{Responsiveness} \rightarrow \text{Vacuum} \rightarrow \text{Physics} \rightarrow \text{Complexity} \rightarrow \text{Observation} \rightarrow (\text{the universe discovers responsiveness}).$

The Boundary of the Spine

the decisive-test scorecard — what responsiveness reaches, hosts, and where it stops



Verdicts verbatim from GRUT_V4_SPINE.md Part IV · A = on-ladder (organized) · B = hosted SM matter-content (off-ladder) · C = observer/meta

Figure 3. What responsiveness reaches, hosts, and cannot reach — the decisive-test scorecard.

Parts I–III walked the ladder from the floor to the observer. Part IV turns to the rungs that are **not yet settled** — the live frontiers where a completed GRUT ToE will eventually have to place an answer, and where v4 stands today with answers that are partial, forbidden, hosted, or open.

This Part is structured so that the *form* enforces the honesty. Each frontier is a chapter titled "**How Does X Emerge in a Responsive Universe?**", and **every chapter answers the same five questions in the same order**:

- **What is known** — observationally, or in standard physics, independent of GRUT.
- **What GRUT already explains** — **DERIVED**. Only results that survive registry audit at the computed/foundational tier.
- **What GRUT constrains** — **FORBIDDEN-by-theorem**. The no-gos; what the framework rules *out*.
- **What GRUT hosts** — **HOSTED / substrate-resident / inputs**. What rides on GRUT's structure or is received into it.
- **What remains unknown** — **OPEN / CONJECTURAL**. The honest negative; the unbuilt.

The five-point format **is the overclaim guard**. Drafting a frontier means filling all five cells — *including when a cell is "nothing."* A frontier whose row-2 (DERIVED) is empty and whose row-5 (OPEN) is full is not a failure of the framework; it is the framework reporting its own boundary at the standing the registry certifies. The reader can audit any claim by reading down the column it sits in.

The decisive test gives Part IV its categories — and the spine its edges. Beyond the five-point gate, four of these frontiers have now been run through a sharper question: *does the sector organize around responsiveness — finding its place on the ladder — or does it require unrelated machinery bolted on from outside?* The answers do not all land the same way, and that is the point. They sort into three kinds:

- **IV•A — On the Ladder.** The sector organizes around responsiveness; its structure descends from (or is forbidden by) the framework's own content. *Dark Matter* (ORGANIZED, substrate- resident in F(t)) and *Quantum Gravity* (PARTIAL — the response is derived, the geometry is hosted) live here.
- **IV•B — Hosted Matter-Content.** Off the ladder: Standard-Model inputs the vacuum *hosts*, not responsiveness questions. The structure is adopted from elsewhere; where the responsive route was tested it pointed *away*. *Neutrinos*, *Baryogenesis*, and *Coupling Unification* are all container-seams — the third with an OPEN, never-attempted seam rather than a tried-and-missed one.
- **IV•C — Observer & Meta Frontiers.** A different kind entirely — neither vacuum-response nor matter-content. The deepest and most speculative: *Consciousness*, *Information*, *Mathematics*. Honestly mostly conjectural or open.

A standing rule governs the richest frontier and recurs throughout: **dark matter is a RESIDENT of GRUT's own pre-responsive substrate sector F(t)** — internal to the layered ontology ($Q \rightarrow F(t) \rightarrow \text{vacuum} \rightarrow \dots$) at the substrate level, **not** external, **not** "hosted from outside," and **not** a successor theory. The phrase "*a different theory, not GRUT*" is reserved exclusively for the vacuum-pole case — a new *propagating vacuum mode* — and is invoked nowhere else.

The boundary is a feature, not a defect. That the frontiers split A / B / C — that some sectors land on the ladder, some are hosted matter-content the vacuum carries but does not organize, and some are observer/meta questions of a wholly different type — is precisely what a discriminating test should produce. A theory that "explained everything" the same way would be explaining nothing in particular. The categories give the spine *edges*: they mark where responsiveness reaches, where it merely hosts, and where the framework is honest that it has not yet reached at all.

The Scorecard

Each frontier, its decisive-test verdict (where run), its category, and one line of ladder placement. The Category-A cosmological results derived earlier — **Dark Energy**, the **hierarchy's existence**, and **$\alpha = 1/3$** — are listed for completeness; they are covered in Parts I–II and are **not** re-derived here.

Frontier	Decisive-test verdict	Category	Ladder placement (one line)
Dark Energy (<i>Parts I–II</i>)	ORGANIZED	A — On the Ladder	Ω_Λ from the responsive vacuum's own α -spine (number anchored 0.6886; tree $(2-R)^2 = 0.71453$, never conflated); covered earlier.
Hierarchy (existence) (<i>Parts I–II</i>)	ORGANIZED	A — On the Ladder	$\tau_0 \gg \tau_{\text{micro}}$ FORCED by the Mori–Zwanzig slow/fast split; covered earlier.
$\alpha = 1/3$ (<i>Parts I–II</i>)	ORGANIZED (value adopted)	A — On the Ladder	The vacuum response coefficient; value adopted, first-principles origin OPEN; covered earlier.
Dark Matter	ORGANIZED (specified)	A — On the Ladder	DERIVED cell empty; the no-gos <i>specify</i> the missing object, hosted substrate-RESIDENT in F(t) as a specified-and-unmet 2-dial extension.
Quantum Gravity	PARTIAL (leaning organized)	A — On the Ladder	Gravity-as-RESPONSE derived ($\Phi_{\mu\nu}$, $\mu_{\text{linear}}=1$, GR-as-limit) + UV no-go internal; gravity-as-GEOMETRY hosted.
Neutrinos / Flavor	CONTAINER-SEAM	B — Hosted Matter-Content	Consequences real ($K=2/3$, NH, $a_v=1$) but the $Z_3/a=\sqrt{2}/M_0$ ansatz is hosted; responsive route misses ($K=4/9$).
Baryogenesis	CONTAINER-SEAM (weaker)	B — Hosted Matter-Content	η_B magnitude is hosted $J_{\text{CP}} \times$ imported K_{neq} ; the $(2-R_B)/S_B$ factor is off-spine and cosmetic.
Coupling Unification	<i>untested; likely CONTAINER-SEAM</i>	B — Hosted Matter-Content	Gauge running + 8.9% miss are pure hosted SM (zero GRUT input); the responsive $\Delta\beta(\alpha_{\text{eff}}(\omega))$ is prose-only/unbuilt. QM/GR meta-unification DERIVED (separate, A).
Consciousness	<i>untested; CONJECTURAL</i>	C — Observer & Meta	The in-in self-comparison hook; a proven structure, no theory of mind. (<i>Also Part III, Ch 10.</i>)
Information	<i>untested through the decisive lens</i>	C — Observer & Meta	Arrow of time DERIVED; BH-information softened at linear order only; mostly OPEN.
Mathematics	<i>untested; conjectural</i>	C — Observer & Meta	No GRUT theorem; a labeled philosophical slot, carrying no registry claim.

A standing rule applies to the verdicts column: **the four decisive-test results (Dark Matter, Quantum Gravity, Neutrinos, Baryogenesis) are authoritative**; the frontiers marked *untested* have **not** been run through the decisive lens and carry **no verdict they have not earned** — they are labeled honestly as *untested/conjectural*, not assigned a category-defining result by analogy.

The Verdict Legend — What Each Word Requires

The verdict words above (ORGANIZED, PARTIAL, CONTAINER-SEAM, UNTESTED) are not free-floating labels. Each is pinned to a **required cell pattern** — which of a frontier's five rows (1 known · 2 DERIVED · 3 FORBIDDEN · 4 HOSTED · 5 OPEN) must be filled — so a badge cannot drift from what the frontier actually shows. This is to the verdicts what the five-tier legend is to the tiers; the distinction it most enforces is **derived vs. specified** — the difference between making a mechanism and merely triangulating a missing one.

Verdict	Required cell pattern	Cat.	Anti-laundering guard
ORGANIZED (derived)	Row 2 (DERIVED) non-empty: the core <i>mechanism</i> is made from GRUT's own structure; any hosting is of a number/anchor, not the mechanism.	A	Badge means "mechanism derived." The cell still shows what is anchored or conditional (Ω _A: mechanism derived / value 0.6886 anchored; α : conditional a/c = 1/3 proven / value adopted / antecedent OPEN).
ORGANIZED (specified)	Row 2 (DERIVED) EMPTY; Row 3 (FORBIDDEN) non-empty and <i>specifying</i> — the no-gos triangulate the missing object; Row 4 hosts it into an <i>internal</i> layer (vacuum or F(t)), never foreign content.	A	Must print "DERIVED cell: EMPTY." Reached by GRUT <i>forbidding</i> / <i>specifying</i> , not <i>making</i> — never readable as a derivation. Free-parameter cost stated (DM: 2 dials, 0-of-3 sourced).
PARTIAL	Make-or-break legs split: ≥ 1 core leg in Row 2 (DERIVED), ≥ 1 core leg in Row 4 (HOSTED).	A	Must name which leg is derived and which hosted (QG: response derived, geometry hosted). The hosted leg keeps full HOSTED standing — no averaging.
CONTAINER-SEAM	Row 2 empty of native content; Row 4 hosts <i>foreign</i> (SM) content; Row 3 empty of any specifying theorem; the responsive route was tried-and-missed (or never run $\rightarrow$ an OPEN seam).	B	"Consequences are real" must always carry "conditional on the hosted ansatz." No consequence promotes the sector off B.
UNTESTED	Five cells filled; the organize-or-host question not yet run.	—	No verdict, and no category by analogy.

IV·A — On the Ladder

The sectors that organize around responsiveness: their structure descends from the framework's own content, or is forbidden by its own theorems. This is where the spine reaches.

Frontier 1 — How Does Dark Matter Emerge in a Responsive Universe?

The richest frontier, and the model for the format. Read the no-gos here not as three separate prohibitions but as a single specification: together they describe the missing object to the dex.

DECISIVE-TEST VERDICT: ORGANIZED (specified) — substrate-resident. DERIVED cell: EMPTY. Dark matter is hosted-by-the-vacuum but **RESIDENT in GRUT's own pre-responsive substrate $F(t)$** — on the ladder, a deeper layer ($Q \rightarrow F(t) \rightarrow \text{vacuum} \rightarrow \dots$). The vacuum's three no-gos hold; $F(t)$ sits outside their jurisdiction because they are theorems about the vacuum's action, not the substrate beneath it. The missing host is a **precisely-specified 2-dial conjectural extension** — sharpened impossibility, not a derivation (GRUT_DM_NONTHERMAL_COHERENT_TEST.md, GRUT_FT_CONDENSATE_PROGRAM.md).

1. What is known. Roughly 85% of the universe's matter is electromagnetically dark, with $\Omega_{\text{dm}} h^2 \approx 0.12$ (Planck). The leading particle-independent class compatible with small-scale structure is wave / fuzzy / coherent (axion-like) dark matter: a single coherent low-momentum scalar field, never thermalized, zero entropy. Its phenomenological window is standard and external — $\log_{10}(m/\text{eV}) \in [-22, -19]$, with kpc-scale soliton cores at the canonical $\sim 10^{-22}$ eV (de Broglie $\lambda = \hbar/mv = 1.92$ kpc at $v = 10$ km/s), and a Lyman- α floor pushing the lower edge toward $\sim 2 \times 10^{-21}$ eV.

2. What GRUT already explains — DERIVED. *Nothing is derived as cold dark matter, and this is stated without flinching.* The only relic GRUT can force — the elastic shear phonon at $k_B T_c = 4.71$ keV, fixed by the world-crystal identity from $\{\hbar, c, \tau_{\text{micro}}\}$ — is **derived-but-REFUTED**: it is warm, has sound speed $c_s = c$ (a stiff relativistic solid with no slow phonon), and overcloses by $\sim 418 \times -5800 \times$. A derived-but-refuted prediction is more scientific than a free dial, and it is banked as such — but the DERIVED cell for *cold* dark matter is, honestly, empty.

3. What GRUT constrains — FORBIDDEN-by-theorem. This is where the frontier is rich. Three theorems about the **vacuum's own action** jointly forbid the vacuum from generating dark matter, and — read together — they *specify* the missing object:

- `vacuum_spectrum_pole_classification` (registry.py:5348, computed) — the responsive vacuum is **single-mode** (a 12,800-pole scan finds no stable off-axis dark-capable pole); dark matter is therefore a hosted matter field, not a vacuum response. (*The single-mode result rests on one channel-counting postulate, honestly flagged — see row 5.*)
- `propagating_relic_forbidden_pincer` (registry.py:5479, conjectural = theorem-modulo-Ostrogradsky-leg) — a propagating dark relic built from the vacuum's action needs a higher-derivative TT operator $\square$ Ostrogradsky ghost $\square \text{Im } \chi < 0 \square N < 0$ by FDT $\square$ violates Q.
- `locality_no_halo_theorem` (registry.py:5540, computed) — a covariant, local, nonlinear vacuum response is no more spatially extended than its baryon source; an extended $1/r^2$ halo needs a $1/k^2$ inverse-Laplacian pole that locality forbids.

Read as a shopping list, the three no-gos specify the 4-item bar the missing object must clear:

non-thermal/coherent (escapes the warm-relic wall), **intermediate scale** (in the empty middle of the τ -gap), **sourced outside the vacuum action** (so it trips no theorem), and **gravitates via ordinary GR, clusters, stable ≥ 14 Gyr, delivers $\Omega_{\text{dm}} h^2 \approx 0.12$** . The registry itself names the remedy in advance: a "FOUNDATIONAL EXTENSION (a microscopic medium with massive excitations)" (registry.py:5500-5501).

4. What GRUT hosts — substrate-resident in $F(t)$. Dark matter is **HOSTED — and the host is internal to GRUT's layered ontology**. It is a no-go-clean coherent scalar resident in GRUT's *own* pre-responsive substrate $F(t)$, the microscopic KMS bath the framework already posits *beneath* the vacuum ("beneath the vacuum itself, lies the microscopic KMS bath $F(t)$: the only place a genuine cold relic could attach"). The three no-gos have **no jurisdiction** over $F(t)$ precisely because they are theorems about the *vacuum's* action, not the substrate beneath it (verified `pole_spectrum.verify()` all-True). The $F(t)$ condensate program works out the minimal host explicitly: a **textbook misalignment scalar** $V(\phi) = \frac{1}{2}m^2\phi^2$, produced by vacuum misalignment, redshifting as a^{-3} once $H \sim m$ (automatically cold, stable $\gg 14$ Gyr), gravitating through the *permitted* Newtonian Green's function — no $1/k^2$ kernel, no second vacuum pole, no new relaxational mode. Its honest cost: **exactly 2 inserted dimensionful dials (the mass m , the amplitude ϕ_i) plus 1 inserted potential shape; 0 of 3 requirements sourced from the bath**. This is GRUT detailing its own deeper substrate layer — *the analogy is that $F(t)$ -DM is to GRUT what the Standard Model is to GR's $T_{\mu\nu}$* : the content of a matter/substrate slot the umbrella theory already has, **not** a successor to it.

Spec — The $F(t)$ condensate host. Tier: CONJECTURAL (no-go-clean). *The minimal host is a textbook misalignment scalar $V(\phi) = \frac{1}{2}m^2\phi^2$, redshifting as a^{-3} once $H \sim m$ (automatically cold, stable $\gg 14$ Gyr), gravitating through the permitted Newtonian Green's function — no $1/k^2$ kernel, no second vacuum pole, no new relaxational mode, so it trips none of the three vacuum no-gos. Honest cost: exactly 2 inserted dimensionful dials (mass m , amplitude ϕ_i) + 1 potential shape; 0 of 3 requirements sourced from the bath. Specified-and-unmet to the dex: the bath's geometric-mean bridge misses the fuzzy-DM window center ($\log_{10} m/\text{eV} = -20.50$) by 7.19 dex. Outside the zero-parameter v4 core. (GRUT_FT_CONDENSATE_PROGRAM.md.)*

5. What remains unknown — CONJECTURAL. The host is **specified-and-unmet to the dex**. The fuzzy-DM window sits in the genuinely empty middle of GRUT's 34-order τ -gap: the two bath anchors span $\log_{10}(\tau_0/\tau_{\text{micro}}) = 33.976$ dex ($E_{\text{micro}} = 4713.68$ eV at $\log_{10} +3.673$; $E_{\tau_0} = 4.978 \times 10^{-31}$ eV at $\log_{10} -30.303$), the window partition closes exactly (22.673 upper + 3.000 window + 8.303 lower = 33.976), and the bath's only natural dimensionless bridge — the geometric mean $\sqrt{E_{\text{micro}} \cdot E_{\tau_0}} = 4.84 \times 10^{-14}$ eV ($\log_{10} = -13.31$) — **misses the window center ($\log_{10} = -20.50$) by 7.19 dex**. The bath supplies neither the mass, nor the VEV (its onset seed is symmetrized KMS noise with $\langle h \rangle = 0$, random phase — it *fluctuates*, it does not *condense*), nor the amplitude ($f(T_c) = \tanh(1/2) = 0.4621$ is dimensionless). **Tier: CONJECTURAL — a deeper-substrate-sector result of GRUT, never a v4-core derivation** (2 free dimensionful parameters keep it outside the zero-free-parameter core).

OVERCLAIM FLAG (Appendix A #1). "GRUT derives / explains dark matter" is the headline overclaim — it is **HOSTED**, received into the substrate. Equally, do not call the $F(t)$ host a "different / successor theory": it is GRUT's own substrate slot, internal to $Q \rightarrow F(t) \rightarrow \text{vacuum}$. **Falsifier:** Lyman- α forest + Milky-Way dwarf-satellite counts pushing the fuzzy-DM mass floor above $\sim 2 \times 10^{-21}$ eV close the kpc-core window; the substrate slot is then empty regardless of what $F(t)$ could supply, and the impossibility becomes total rather than sharpened.

Frontier 4 — How Does Quantum Gravity Emerge in a Responsive Universe?

The medium, the UV no-go, and the hierarchy — the boundary of GRUT's explanatory reach, drawn sharply.

DECISIVE-TEST VERDICT: PARTIAL (leaning organized). *The two make-or-break legs split. Gravity-as-RESPONSE is DERIVED* ($\Phi_{\mu\nu} = \alpha \hat{\chi} \cdot P^{TT}$, $\hat{\chi}$ the normalized memory pole; the μ -linear = 1 projector theorem; GR as a limit) **and the UV-frontier is a framework-internal no-go** — both organized. But **gravity-as-GEOMETRY is HOSTED**: the Einstein tensor $G_{\mu\nu}$, the metric $g_{\mu\nu}$, and the Einstein–Hilbert term enter as inputs, not as outputs of responsiveness. GRUT modifies GR with a response; it does not derive geometry. "Derives how gravity responds, not why it is geometric." The strongest Category-A cell short of dark matter — but qualified, not clean (GRUT_QG_RESPONSIVENESS_TEST.md).

1. What is known. QM and GR must be reconciled; there is a Planck scale; and there is a hierarchy problem — why microscopic and macroscopic scales are so vastly separated.

2. What GRUT already explains — DERIVED. Unification is DERIVED as a single parameter space: QM is the $\tau_0 \rightarrow 0$ limit and GR is the $\omega\tau_0 \rightarrow \infty$ limit of **one** CTP constitutive action, both limits exact, with the crossover the crystalline boundary $X = \Lambda_{\text{grav}} \cdot \tau_0 = 1$. The unification scale is the ~ 689 Hz table-top decoherence plateau, **not** the Planck energy — which is why GRUT has near-term falsifiers at all. **The decisive-test "response" leg lives here:** $\Phi_{\mu\nu} = \alpha \hat{\chi}(\omega) \cdot P^{TT}$ is derived by varying the constitutive cross term (the normalized memory pole $\hat{\chi}(\omega) = 1/(1 - i\omega\tau_0)$ is the constitutive response — responsiveness itself; the full susceptibility is $\chi = \alpha \hat{\chi} = \alpha/(1 - i\omega\tau_0)$); **μ -linear = 1** is a theorem forced by the transverse-tracefree projector's own index structure (the ladder rules out its own would-be scalar modification, over-determined $\sim 32\sigma$ by the CMB-ISW); and GR is recovered as the high- ω limit ($n_g(0) = \sqrt{4/3}$, $n_g(\infty) = 1$) — all computed-tier and verified. The **existence** of the hierarchy is FORCED: the Mori–Zwanzig slow/fast projection requires $\tau_0 \gg \tau_{\text{micro}}$, and the no-dark-mode result is magnitude-inert across 34 orders, flipping only at the pure ratio $\tau_K/\tau_0 = 1/4$ (GRUT_HIERARCHY_LEDGER.md §§2C, 3).

3. What GRUT constrains — FORBIDDEN-by-theorem. The hierarchy *magnitude* is FORBIDDEN: there is no β -function that carries τ_0 , and forcing the gap would require a new propagating vacuum pole the framework's minimalism forbids (GRUT_HIERARCHY_LEDGER.md §5). The conformal mode σ — the natural UV carrier — is caught in the **locality–dynamics fork** (GRUT_UV_SECTOR_CONSTRUCTION.md §5): the anomaly number that would source a Riegert action lives in the E_4 / a-channel, which GRUT's own second-order analysis proves is Lovelock-null and dynamically dormant; the only *live* channel is c / Weyl^2 . So σ is either locality-safe and ghost-free but **inert** (confined to E_4 , no β -function), or dynamically live but then it is the metric trace tied to $T_{\mu\nu}$ by the Ward identity — giving a $1/k^4$ matter→metric pole (strictly worse than the forbidden $1/k^2$, verified as a divergent limit) **and** a fourth-order Ostrogradsky ghost (opposite-sign residues $\square \text{Im } \chi < 0 \square N < 0 \square$ non-unitary). No window exists where σ is simultaneously locality-safe, Q-safe, and live — a **structural NO**. *This is the decisive-test's second organized leg: the UV obstruction is a registry computed theorem descending from the CTP action, not a post-hoc alibi — the ladder setting its own UV edge.* **The decisive unity: the dark-matter no-go and the hierarchy-magnitude no-go are the SAME no-go** — both forbid a new propagating vacuum pole.

Spec — The unified no-go. Tier: FORBIDDEN-by-theorem. The dark-matter no-go and the hierarchy-magnitude no-go are the **same** no-go: both forbid a new propagating vacuum pole, from the vacuum's own action, by **locality + Q**. How: such a pole needs a higher-derivative TT operator $\square$ Ostrogradsky ghost $\square \text{Im } \chi < 0 \square$ noise $N < 0$ by FDT $\square$ violates Q; equally, an extended $1/r^2$ halo needs a $1/k^2$ inverse-Laplacian pole a local kernel cannot contain. The same minimalism that unifies QM and GR as two exact limits makes both unknowns structural rather than accidental.
(propagating_relic_forbidden_pincer, registry.py:5479; GRUT_HIERARCHY_LEDGER.md §5.)

4. What GRUT hosts — HOSTED. The geometry is hosted — this is the seam the decisive test names. In the derivation the linearized Einstein tensor is a literal placeholder symbol ($G_{\text{einstein_one}} = \text{sp.Symbol}("G_{\text{einstein_1}}")$), the EoM reads $G_{\mu\nu}^{(1)} - \Phi_{\mu\nu} = 8\pi G T_{\mu\nu}$ with $G_{\mu\nu}^{(1)}$ on the *input* side, the Einstein–Hilbert coefficient is a free abstract symbol, and the influence-functional core carries no metric/curvature content at all — $g_{\mu\nu}$ is *set in*, the Einstein tensor is *given*. The would-be deeper account (the elastic-medium "why gravity is geometric") is explicitly **demoted** — an unpaid postulate that predicts a refuted warm dark sector, and is out of the registry. Separately, τ_0 and $\tau_{\text{micro}} (\equiv \hbar/k_B T_c)$ enter as the **two independent anchored numbers** the entire 34-order gap reduces to: $\ln(\tau_0/t_P) = 134.45$ and $\ln(\tau_{\text{micro}}/t_P) = 56.21$. The famous gap $c = \ln(\tau_0/\tau_{\text{micro}}) = 78.23$ is exactly their *difference* ($134.447 - 56.214$, to machine precision), **not** a third number; everything else hierarchy- flavored ($G_0, \ell_{\text{micro}}, L_0, T_{\text{peak}}, 4.71 \text{ keV}, \tau_{\Lambda}, H_0, \Omega_{\Lambda}, \mu_0, a_0, f(T)$) is a disguise of these two (GRUT_HIERARCHY_LEDGER.md §4). (Note: τ_{micro} 's *registry* tier is conjectural — $\tau_{\text{micro_thermal_scale}}$, registry.py:1442 — and is reported in-narrative as hosted-with-forbidden-relation-to- τ_0 .) The two-anchor split is **structurally protected** by the directional RG firewall (the UV cannot reach down to regenerate the IR τ_0 operator), not a loose end.

Spec — Quantum gravity: response derived, geometry hosted. Tier: PARTIAL. The two make-or-break legs split. **Gravity-as-RESPONSE is DERIVED:** $\Phi_{\mu\nu} = \alpha \cdot \hat{\chi}(\omega) \cdot P^{TT}(\hat{\chi})$ (the normalized memory pole) from varying the constitutive cross term, $\mu_{\text{linear}} = 1$, and GR as the high- ω limit ($n_g: \sqrt{4/3} \rightarrow 1$). **But gravity-as-GEOMETRY is HOSTED:** the Einstein tensor $G_{\mu\nu}$, the metric $g_{\mu\nu}$, and the Einstein–Hilbert term are inputs — in code the linearized Einstein tensor is a literal placeholder symbol. How (the seam): GRUT modifies GR with a response; it does not derive why gravity is geometric. Closing that — currently with the elastic-medium candidate demoted — would upgrade **PARTIAL** $\rightarrow$ **ORGANIZED**. (GRUT_QG_RESPONSIVENESS_TEST.md.)

5. What remains unknown — OPEN. No UV-complete *dynamical* sector exists. Deriving $\ln(\tau_{\text{micro}}/t_P) = 56.21$ is "not even askable": GRUT has no condensing scale-invariant potential, no GRUT-origin running coupling, and no Planck-scale boundary condition (GRUT_UV_ANCHOR_FEASIBILITY.md, verdict NOT-YET-FORMULATED). The reference class generically makes large Planck-logs from $O(1)$ inputs (with the genuine $1/16\pi^2$ normalization, $b_0 \approx 0.0063$ gives logs of tens-to-hundreds), but **GRUT's own structure cannot supply the live coupling to realize one** — the class makes the logs, the engine is missing (GRUT_UV_SECTOR_CONSTRUCTION.md §5.1). And the deepest open seam the decisive test isolates: **a derivation of the geometric/ curvature structure itself from responsiveness** — currently hosted, with the elastic-medium candidate demoted; closing it would upgrade **PARTIAL** $\rightarrow$ **ORGANIZED**.

OVERCLAIM FLAGS (Appendix A #2, #6, #16). The hierarchy magnitude is the single fact that sinks the spine if inflated — booked FORBIDDEN+anchored, never "derived." $\alpha = 1/3$'s value is adopted; its first-principles (Riegert/IR-carrier) derivation is OPEN. G_0 's value is derived from $\{\hbar, c, \tau_{\text{micro}}\}$ but the viscoelastic-SOLID premise is an UNPAID POSTULATE (one of ≥ 4 degenerate medium classes). The two-anchor structure is **protected, not a gap**. **Per the decisive test: derive the response of gravity, do not claim the geometry of gravity** — the Einstein tensor is hosted.

IV·B — Hosted Matter-Content

Off the ladder — Standard-Model inputs the vacuum hosts, not responsiveness questions. The structure is adopted from elsewhere; where the responsive route was tested it returned the wrong number and pointed away from the spine. The container carries the right numbers; it does not generate them.

Frontier 2 — How Do Neutrinos Emerge in a Responsive Universe?

DECISIVE-TEST VERDICT: CONTAINER-SEAM. *The consequences are real and machine-verified (Koide $K = 2/3$, normal-ordering preference, $a_v = 1$) — but the structure that produces them (the Z_3 circulant, $a = \sqrt{2}$, M_0) is an **adopted Yukawa ansatz GRUT hosts, not generates**. The responsive (CTP fixed-point) route was tested and **MISSES**: it returns $K = 4/9$, not $2/3$ — actively pointing toward external Yukawa input. Off-ladder (GRUT_NEUTRINO_RESPONSIVENESS_TEST.md).*

1. What is known. Neutrinos oscillate, so they are massive; the mass-splitting ratio $\Delta m^2_{\text{atm}}/\Delta m^2_{\text{sol}} \approx 33.9$; the charged-lepton masses satisfy the Koide relation $K = 2/3$ to high precision; the mass ordering (normal vs inverted) is undetermined; Σm_ν is bounded by cosmology, and the Dirac-vs-Majorana question is open pending neutrinoless double-beta decay.

2. What GRUT already explains — DERIVED [SPLIT]. The charged-lepton **$K = 2/3$ holds at machine precision as an algebraic identity** of a Z_3 -circulant mass operator, for any overall scale M_0 and any phase θ (koide_z3_circulant_structure, registry.py:2808, computed) — DERIVED as an *identity*, not a mechanism. Conditional on the structural coupling **$a_v = 1$** , the generalized ansatz admits a unique normal-hierarchy interior solution ($\Sigma m_\nu \approx 60$ meV, $m_\beta \approx 9$ meV), so GRUT **prefers normal ordering** — and $a_v = 1$ is itself DERIVED as the unique boundary-degenerate Z_3 coupling (neutrino_z3_coupling_a_equals_1_uniqueness_theorem, registry.py:2983, computed; Correction #29 — no longer a postulate). The NH preference and $\Sigma m_\nu \approx 60$ meV are anchored-on the $a_v = 1$ derivation (neutrino_hierarchy_z3_nh_prediction, registry.py:2926, **anchored** — tier and notes consistent). *Per the decisive test, every one of these is a consequence conditional on the adopted Z_3 ansatz, not a descent from the ladder.*

Spec — Koide $K = 2/3$ and normal ordering. Tier: DERIVED [SPLIT]. *The charged-lepton $K = 2/3$ holds at machine precision as an algebraic identity of a Z_3 -circulant mass operator (any scale M_0 , any phase θ). Conditional on the structural coupling $a_v = 1$ — itself derived as the unique boundary-degenerate Z_3 coupling — the ansatz admits a unique normal-hierarchy solution ($\Sigma m_\nu \approx 60$ meV, $m_\beta \approx 9$ meV), so GRUT **prefers normal ordering**. **Split:** these are consequences of an adopted Z_3 ansatz GRUT hosts, not generates — and the responsive (CTP fixed-point) route, where tested, returns $K = 4/9$, missing $2/3$. (koide_z3_circulant_structure, computed, registry.py:2808.)*

3. What GRUT constrains — FORBIDDEN-by-computation. The charged-lepton Z_3 structure provably does **not** extend to neutrinos: the same $a = \sqrt{2}$ ansatz that yields $K = 2/3$ forces $\Delta m^2_{\text{atm}}/\Delta m^2_{\text{sol}} \geq 194.1$, against the observed 33.9 — a factor of ~ 6 too large, computed and unconditional (charged_lepton_z3_does_not_extend_to_neutrinos, registry.py:2882, computed). This is a virtue, not a gap: **the wall is the physics** — the layered ontology showing as an internal computation that *forbids* the naive extension and forces a distinct neutrino coupling ($K_v = 1/2$, $a_v = 1$).

4. What GRUT hosts — HOSTED. The Standard-Model lepton field content and the existence of the neutrino sector are received, not generated. The circulant parameters M_0 and θ are *fitted* to the masses — the Z_3 circulant is a parameterization, hosted as structure. *The decisive test sharpens this: the one tie to responsiveness, $\theta = 2/9 = K \cdot \alpha_{\text{vac}}$, is statistically real but mechanistically decorative, and collapses to a single α_{vac} via $K \equiv 2 \cdot \alpha_{\text{vac}}$; and M_0 is a hard free fit, native $\mu_0 = \hbar/\tau_0 \approx 5 \times 10^{-31}$ eV sitting ~ 20 orders below the fitted $0.560 \text{ GeV}^{1/2}$.*

5. What remains unknown — OPEN. The flavor *mechanism* — what physically selects the Z_3 structure, M_0 , and θ from the CTP fixed point — is OPEN (koide_phase_4_open_negative); so is the deeper KS-anomaly account of *why* neutrinos lack the EM channel that gives charged leptons $a = \sqrt{2}$. The responsive candidate (the multi-flavor CTP fixed point, Conjecture F1) is a HYPOTHESIS that, where tested quantitatively, fails ($K = 4/9$). GRUT's Dirac preference is a closeness preference ($a/c = 1.15525$ vs Majorana 1.17256 , against canonical $\sqrt{4/3} = 1.15470$), not a proof (neutrino_dirac_prediction, registry.py:2498, anchored — OPEN remainder).

OVERCLAIM FLAGS (Appendix A #9, #11). "GRUT forces Z_3 " — no; the circulant is assumed, and " $K = 2/3$ is a theorem of Z_3 " is the honest claim. The normal-ordering preference is DERIVED-from- $\alpha_v=1$, not parameter-free omniscience. **Live falsifiers:** a positive $\nu\beta\beta$ signal kills Dirac; JUNO/DUNE confirming inverted ordering, or DESI/Euclid pushing Σm_ν out of band, kills the NH preference. **Upgrade to ORGANIZED:** derive the Z_3 Jacobian + $a = \sqrt{2}$ from the CTP fixed point (yielding $2/3$, not $4/9$), or anchor M_0 — either would move the sector onto the ladder.

Frontier 3 — How Does Baryogenesis Emerge in a Responsive Universe?

DECISIVE-TEST VERDICT: CONTAINER-SEAM (weaker than flavor) + corrected overclaim. The η_B magnitude is **hosted SM Jarlskog $J_{\text{CP}} \times \text{imported/asserted } K_{\text{neq}}$** ; the GRUT factor $(2 - R_B)/S_B$ is **off-spine and cosmetic** — $R_B = 1.018 \neq$ the canonical $R = \sqrt{4/3}$, $S_B = 4\pi \cdot 45 \neq S = 108\pi$, its numerator $(2 - R_B)$ is $O(1)$ (0.982) and the full factor moves η_B by $<2\%$, and the $+7.7\%$ match was produced by a documented S_B re-choice (a **reverse-fit**; using the canonical R worsens the fit to -7.3%). Re-tiered **ANCHORED**. Off-ladder (GRUT_BARYOGENESIS_RESPONSIVENESS_TEST.md).

1. What is known. The baryon-to-photon ratio is $\eta_B \approx 6.1 \times 10^{-10}$ (Planck) — a small, nonzero matter–antimatter asymmetry requiring the three Sakharov conditions.

2. What GRUT already explains — DERIVED [SPLIT]. The *existence* of a nonzero asymmetry is derived: the CTP path is intrinsically asymmetric (the canonical ratio $R \neq 1$), and the Sakharov CP-content is forced by the in-in asymmetry — this is the derived half. *The decisive test corrects the prior framing here: the native structural factor $(2 - R_B)/S_B$ is NOT the canonical α -spine and is NOT load-bearing — its numerator $(2 - R_B)$ is an $O(1)$ dressing ($=0.982$; the full factor $\approx 1.7 \times 10^{-3}$), and the registry's "asymmetry vanishes at $R = 1$ " reading is misleading (the algebraic zero of $(2 - R)$ is at $R = 2$; across the whole R_B range the factor moves η_B by $<2\%$).* The genuinely on-spine remnant is the narrative — the constitutive lag supplying non-equilibrium across the EW crossover, a real responsiveness mechanism for Sakharov condition (3) — but it is only a narrative; the number is not derived. The baryogenesis_eta_b claim (registry.py:1696) is re-tiered **ANCHORED**, not a zero-parameter computed prediction.

3. What GRUT constrains — FORBIDDEN-by-theorem. Nothing additional is forbidden at this frontier beyond the general in-in arrow that makes $R \neq 1$ mandatory; the CP-violating direction is *forced* by Q's path

asymmetry rather than ruled out. (Cell honestly thin.)

4. What GRUT hosts — HOSTED. The Jarlskog invariant J_{CP} and the electroweak-nonequilibrium factor K_{neq} are Standard-Model inputs, received not generated. **The magnitude rides entirely on these hosted numbers** — J_{CP} ($\approx 3.18 \times 10^{-5}$, SM/CKM) $\times K_{neq}$ ($\approx 1.19 \times 10^{-2}$, imported, hardcoded, $\pm 50\%$ structural, *uncomputed in code* despite the doc's $\alpha_{eff} \times \delta T/T \times v/T$ claim) / S_B — the smallness 6.7×10^{-10} is set *before* the GRUT factor touches it. The $(2-R_B)/S_B$ factor is off-spine and cosmetic, as the decisive test establishes.

5. What remains unknown — OPEN. The *magnitude mechanism* is not zero-parameter and is, on the decisive test, **assembled-to-fit**: R_B is route-ambiguous to $18,000 \times$ (Route 1 = 1.018 vs Route 2 = 5.6×10^{-5} , the matching route selected), K_{neq} is a $\pm 50\%$ uncomputed knob, and the canonical spine R is *not* used because it degrades the fit. GRUT's route lands $\eta_B \approx 6.57 \times 10^{-10}$, about $+7.7\%$ above Planck — an agreement assisted by hosted inputs and produced by the S_B re-choice, not a clean prediction.

OVERCLAIM FLAG (Appendix A #4). "GRUT predicts $\eta_B = 6.6 \times 10^{-10}$ " overclaims — **existence is DERIVED, magnitude is HOSTED** ($J_{CP} \times$ imported K_{neq}) and the GRUT factor is off-spine and cosmetic; the $+7.7\%$ match was reverse-fit via a documented S_B re-choice and must not be read as zero-parameter. **Upgrade to PARTIAL/ORGANIZED:** derive K_{neq} from the constitutive lag in code **and** show the load-bearing factor uses the canonical $R = \sqrt{4/3}$, $S = 108\pi$ and still lands η_B (currently fails by construction: canonical R gives -7.3%).

Frontier 5 — How Does Coupling Unification Emerge in a Responsive Universe?

The meta-claim, and its honest status.

DECISIVE-TEST STATUS: UNTESTED through the lens — likely CONTAINER-SEAM, Category B (no verdict asserted). The gauge couplings, β -functions, and the 8.9% unification miss are 100% hosted SM — reproduced in .venv from pure standard-SM inputs (Machacek–Vaughn β , PDG couplings) with **zero GRUT input**. The proposed responsive correction $\Delta\beta(\alpha_{eff}(\omega))$ is prose-only vaporware: a named hook ($K(t) \rightarrow \delta\beta_i$) but no equation, no tie to $\chi(\omega)$ in the gauge sector, no derivation, no closure — a conjecture **never attempted** (OPEN, not tried-and-missed). The QM/GR meta-unification is **DERIVED** (separate, Category A). Registry track_v_coupling_unification_open_question is honestly open_negative. See theory/GRUT_COUPLING_UNIFICATION_RESPONSIVENESS_TEST.md.

1. What is known. The three Standard-Model gauge couplings run toward near-coincidence at $\sim 10^{16}$ GeV but miss exact unification by $\sim 8.9\%$ in the canonical (non-SUSY) framework.

2. What GRUT already explains — DERIVED [SPLIT]. The derived object here is the **QM/GR meta-unification, NOT gauge-coupling unification** (gauge-coupling closure is NOT in this cell — see row 5). Unification as a meta-statement is the real content: one CTP action yields the sectoral limits (QM as $\tau_0 \rightarrow 0$, GR as $\omega\tau_0 \rightarrow \infty$), giving a coherent tiered *place* for every sector — not a reconciliation of two languages but a single parameter space. This rides on closed_universe (foundational) and ctp_action_structure (computed) and is DERIVED at the meta-layer.

3. What GRUT constrains — FORBIDDEN-by-theorem. The same minimalism that unifies QM and GR — one dynamical metric, one memory pole — is what *forbids* a second propagating pole. Unification and the no-gos are one fact seen from two sides: the framework cannot buy gauge-coupling closure by adding the kind of new

structure it has already forbidden for dark matter and the hierarchy.

4. What GRUT hosts — HOSTED. The SM gauge group $SU(3) \times SU(2) \times U(1)$ and the running couplings are inputs, received not generated. *(This hosting of the SM gauge content is why the frontier is likely Category B — but the decisive organize-or-host test has not been run, so the category is flagged as likely, not asserted.)*

5. What remains unknown — OPEN. The proposed constitutive β -correction $\Delta\beta(\alpha_{\text{eff}}(\omega))$ that would close the 8.9% gap is **not derived from the CTP action** ($\delta S_{\text{CTP}}/\delta g_i$ has not been computed to yield it). Track V is registered **open_negative** (`track_v_coupling_unification_open_question`, `registry.py:2718`, `open_negative`): the SM couplings miss exact unification by 8.9% at the GUT scale and the constitutive correction is unbuilt.

OVERCLAIM FLAG. *"GRUT unifies the couplings" overclaims — the **meta-unification** (QM/GR as exact limits) is the genuine content; the **gauge-coupling closure is OPEN/unbuilt** (Track V, `open_negative`). Mostly OPEN. No decisive-test verdict is asserted: the sector is untested through that lens.*

IV·C — Observer & Meta Frontiers

A different kind of frontier entirely — neither vacuum-response (IV·A) nor matter-content (IV·B). These are the deepest and most speculative rungs: questions about the observer, about information, and about why the universe is describable at all. None has been run through the decisive organize-or-host test, and none should be assigned a verdict it has not earned. They are reported honestly as conjectural or open.

Frontier 6 — How Does Consciousness Emerge in a Responsive Universe?

The in-in hook — the most speculative frontier, tagged loudly. (Also treated narratively in Part III, Chapter 10 — "Emergence of Mind"; cross-referenced here, not duplicated.)

DECISIVE-TEST STATUS: CONJECTURAL — not run through the lens. The decisive test is a spine-membership test for physics sectors; consciousness is an observer/meta frontier and is **not** assigned an organize-or-host verdict. What GRUT offers is **the in-in self-comparison hook — a proven structure, not a theory of mind.**

1. **What is known.** The observer problem (who collapses the wavefunction); the hard problem of experience; an empirical ~40 Hz neural resonance associated with conscious binding.
2. **What GRUT already explains — DERIVED.** Only the structural object: the CTP **in-in self-comparison**, $S_IF[\varphi_+ = \varphi_-] = 0$ — a closed universe measures itself by maintaining two copies of its history and responding to their *difference*; the influence action vanishes when nothing has changed (`closed_universe`, `registry.py:109`, foundational). This object is *proven*. The 40 Hz *numbers* are computed from two independent framework routes (Λ_{grav} at tubulin-dimer parameters $\rightarrow$ 39.9 Hz; self-referential fixed-point dynamics $\rightarrow$ 41.7 Hz, sharing no fitted parameters).

Spec — The consciousness hook (and only the hook). Tier: structure DERIVED, reading CONJECTURAL [SPLIT]. GRUT contributes one proven object: the CTP **in-in self-comparison**, $S_IF[\varphi_+ = \varphi_-] = 0$ — a closed universe measures itself by maintaining two copies of its history and responding to their difference. The 40 Hz numbers are computed from two independent routes (39.9 Hz, 41.7 Hz; no shared fits). How (the seam): the structure is a theorem; reading it as "the universe knowing itself," or the 40 Hz coincidence as consciousness, is a conjectural interpretation layered on top — **no** account of experience, qualia, or awareness, and nothing in the framework depends on it. (`neural_resonance_speculative`, `conjectural`, `registry.py:2170`.)

3. **What GRUT constrains — FORBIDDEN-by-theorem.** Nothing. (Cell honestly empty.)
4. **What GRUT hosts — HOSTED.** Nothing — no observer postulate is added to the framework. (Cell honestly empty.)
5. **What remains unknown — CONJECTURAL.** The entire **consciousness bridge** — any connection from the fixed-point self-reference to a first-person observable — is unbuilt. Reading the in-in self-comparison as "the universe knowing itself," or the 40 Hz coincidence as consciousness, is a CONJECTURAL interpretation layered on a proven structure (`neural_resonance_speculative`, `registry.py:2170`, `conjectural`).

OVERCLAIM FLAG (Appendix A #5). "GRUT explains consciousness / the universe is conscious" is the cardinal overclaim. This is **a hook and only a hook** — self-reference plus a non-load-bearing numerical coincidence; **no account of experience, qualia, or awareness** exists or is claimed. Nothing in the framework depends on it; if it is wrong, nothing else changes; the v3 reader edition makes no consciousness claim at all. The most speculative frontier — tagged the loudest.

Frontier 7 — How Does Information Emerge in a Responsive Universe?

DECISIVE-TEST STATUS: UNTESTED through the decisive lens. An observer/meta frontier, not run through the organize-or-host test; carries no verdict it has not earned. Independently settled: the arrow of time is *DERIVED*; the black-hole-information result holds at linear order only; the frontier is mostly *OPEN*.

- 1. What is known.** Unitarity / information conservation; the black-hole information puzzle; the Page curve.
- 2. What GRUT already explains — DERIVED.** The **arrow of time** follows from Q's retarded kernel: constitutive evolution carries entropy production $\dot{S} = (1/\tau_0)\langle(z - z_{\text{target}})^2\rangle \geq 0$, zero only at the fixed point, verified strictly positive over 10^5 random states (arrow_of_time_from_entropy, registry.py:1734, computed). **Information conservation** is the in-in diagonal vanishing ($S_{\text{IF}}[\varphi_+ = \varphi_-] = 0$, part of Q), and the memory horizon is $L_0 = c\tau_0$.
- 3. What GRUT constrains — FORBIDDEN-by-theorem.** The FDT positivity $N \geq 0$ forbids wrong-sign relics — an information-positivity selection rule (the same $N < 0$ violation that kills the propagating dark relic and the Ostrogradsky ghost).
- 4. What GRUT hosts — HOSTED / anchored.** The partial black-hole-information / Page-curve result holds at linear order and is received at that standing (bh_information_partial, registry.py:1773, anchored).
- 5. What remains unknown — OPEN.** A fully nonlinear, covariant account of black-hole evaporation is not closed; GRUT softens the puzzle only at the linearized level.

OVERCLAIM FLAG. GRUT softens the BH information puzzle **at linearized order only** — do not claim a resolution. Mostly *OPEN*.

Frontier 8 — How Does Mathematics Emerge in a Responsive Universe?

Honestly mostly-OPEN — the slot, placed for map-completeness.

DECISIVE-TEST STATUS: UNTESTED / conjectural. Not run through the lens, and carrying no registry claim to test — a labeled philosophical slot, reported as such.

- 1. What is known.** The "unreasonable effectiveness" question — why the universe is describable by mathematics at all.
- 2. What GRUT already explains — DERIVED.** Nothing. There is no GRUT theorem about mathematics. (Cell honestly empty.)

3. What GRUT constrains — FORBIDDEN-by-theorem. Nothing. (Cell honestly empty.)

4. What GRUT hosts. The formal apparatus itself — the CTP action, the operator algebra — is *presupposed*, not generated. This is a narrative/philosophical placement carrying **no registry claim** (no machine-checked claim id backs a "mathematics is hosted" statement); it is noted only to keep the cell honest, not to attach a substantive tier.

5. What remains unknown — OPEN / CONJECTURAL. Entirely open. At most a philosophical reflection connects to the reflexive-closure framing of Part III (responsive systems modeling responsiveness; `closed_universe`, `registry.py:109`, `foundational`) — but no claim is made.

OVERCLAIM FLAG. *Written as a labeled open/philosophical entry; **no claims**. This frontier exists to demonstrate the discipline: a sector with four near-empty cells is reported as such, not dressed up.*

And whatever else survives audit. The frontiers above are not a closed list, and the three categories are not a closed taxonomy. Any future sector — a strong-CP / axion entry (currently absent, not forbidden), the tensor-to-scalar ratio r (OPEN, no inflationary epoch), the primordial amplitude A_s (OPEN honest negative), or any puzzle not yet named — enters Part IV by the **same five-point gate**: state what is known, what GRUT derives, what it forbids, what it hosts, and what remains open. And any frontier may be put to the **same decisive test** — does it organize around responsiveness (A), is it matter-content the vacuum hosts (B), or is it an observer/meta question of a different kind (C)? The format is the entry requirement, the decisive test is the category gate, and both are the overclaim guard. A frontier earns a place on the ladder not by being solved, but by accepting honest tiering across all five cells — and by being honest about which side of the spine's edge it falls on.

PART V — THE EMERGENCE OF EVERYTHING

Ladder rungs: all of them, retold once, as a single story. This is the closing volume — synthesis, not derivation. Nothing new is claimed here that has not been earned and tiered in Parts I–IV; what is new is only the act of standing back far enough to see that they are one motion. The spine of this Part is the spine of the whole document:

$Q \rightarrow F(t) \rightarrow \text{Responsiveness} \rightarrow \text{Vacuum} \rightarrow \text{Physics} \rightarrow \text{Complexity} \rightarrow \text{Observation} \rightarrow (\text{the universe discovers responsiveness}).$

Chapter 12 — The Ladder, Told Once

Begin at the bottom, where there is not yet a bottom to speak of — no time, no scale, no length to measure against.

There is **Q**. Not a force, not a field, not an epoch: a structural fact about how anything responds to anything. A system answers its past and never its future; the in-in self-comparison of a closed history vanishes on its own diagonal. This is the one thing the framework does not have to build, because it is a property of the formalism itself — proven, scale-free, prior to everything. Q is the floor that does not rest on a floor. Every later rung is, in the end, Q wearing more structure.

From Q comes **F(t)** — the pre-responsive substrate, the microscopic bath beneath the vacuum that GRUT's own ontology already posits before any vacuum has formed. Here there is matter-of-fact thermal motion but no memory worth the name: a fast bath, churning, forgetting. The honest tier is split (Appendix A #20): the *single-pole form* the memory will eventually take is derived as the Mori–Zwanzig Markovian limit, but F as a standing pillar is POSTULATED — and the value of the relaxation time it will acquire is anchored, not made. The ladder does not pretend otherwise.

From F(t) comes **responsiveness**. This is the genesis event, and it is the one place where the universe stops merely happening and starts *remembering*. When the bath cools through T_c , the fast modes project out and one slow channel fails to be rejected — the un-rejected Mori–Zwanzig remainder becomes a memory. A single proper length $L_0 = c\tau_0$ condenses out of a scale-free substrate, breaking the boundary charge of dilatation invariance (Appendix A #21). With one length and one memory time, the universe has — for the first time — a *now* distinct from a *before*. And the keystone of the whole genesis is qualitative and exact: **Q is the one invariant that survives the loss of memory**. Everything else about the pre-responsive phase is washed out in the crossover; the causal arrow is not, because it was never a feature of the memory in the first place. Time, in the sense that has a direction, is born here — but the *arrow* predates it.

From responsiveness comes the **vacuum**. Once there is one memory channel, there is a memory kernel $K(t) = (1/\tau_0) e^{-t/\tau_0} \Theta(t)$ and a constitutive law $\tau_0 \dot{z} + z = z_{\text{target}}$ — the single load-bearing hub every later sector turns out to be a regime of. The responsive vacuum is the first genuinely *physical* object: it is local, it carries conserved stress-energy through the diffeomorphism Ward identity, and it answers causally and passively because the CTP variation can yield nothing else. This is why responsiveness *looks like a field theory* — not because a field theory was assumed, but because the form-class is forced.

From the vacuum comes **physics**, and this is the widest rung. The known sectors are not separate theories bolted together; they are regimes of the one law. Quantum mechanics is the zero-memory face ($\tau \rightarrow 0$), recovered rather than re-postulated, complete with decoherence, a pointer basis, and the arrow $\dot{S} \geq 0$ — with the single razor-sharp seam that the Born *weights* are received, not derived (Appendix A #14; not a GRUT-specific gap). Gravity is the elastic/curvature response, geometric because L_0 is the one proper length breaking scale-freedom — though the viscoelastic-SOLID premise that underwrites it is an unpaid postulate, one of several degenerate medium classes (Appendix A #16). The cosmos is the law at the largest scale: linear cosmology comes back as Λ CDM *exactly* — $\mu_{\text{linear}} = 1$, the cleanest theorem in the whole framework, α -free and τ_0 -free, over-determined by roughly 32σ — while the dark-energy *number* is honestly anchored (the tree value $(2-R)^2 = 0.71453$ is *not* the anchored full-Friedmann 0.6886; Appendix A #3), and A_s and n_s stay open or fitted. Physics is the rung where the most is *derived* and, precisely because of that, the rung where the honest negatives must be stated loudest.

From physics comes **complexity** — and here the ladder changes character, and the discipline with it. Atoms, stars, galaxies, the heavy elements, the first information-bearing systems: these run on GRUT's recovered- Λ CDM

background as standard astrophysics, hosted and GRUT-consistent, not GRUT-derived. Life, mind, intelligence sit higher still, as *placed slots* — a completed Theory of Everything must eventually have a rung for them, so the rung is named; but biology is an open slot whose only GRUT hook is the derived arrow $\dot{S} \geq 0$ as the precondition for dissipative organization, and consciousness is a conjectural reading of the in-in self-comparison, a hook and not a theory of mind (Appendix A #5). Nothing in the framework depends on these readings, and the reader edition makes no claim on them. The ladder reaches up to them honestly; it does not pretend to have climbed them.

And from complexity come **observers** — the reflexive closure. The same constitutive law that built the cosmos now describes the things inside it that *measure*, because an observer is not an exception to the medium but a crystallized piece of it. A closed universe measures itself by maintaining two copies of its history and responding to their difference — which is just Q, the in-in arrow, read at the top of the ladder instead of the bottom. The motion completes: the universe that began as a causal arrow with no memory ends as a region of medium responsive enough to model the medium, and so to discover that it is responsive at all. **The universe becomes aware of its own history.** That last clause is the softest rung and is labeled as such — aspirational, philosophical, not a derivation — but it is the natural shape of the ladder's own top.

Chapter 13 — Why "Grand Responsive Universe Theory" Earns Its Name

For most of its development the name was an ambition wearing the clothes of a claim. What changed is not that every frontier was solved — it was not — but that the *whole* came to be a single tiered map with a sharp boundary, drawn at exactly the places GRUT's own structure draws it and nowhere else. That is the only sense of "Theory of Everything" that survives a referee: not a derivation of everything, but a coherent **place** for everything, with completeness understood as completeness *of the map* — and that completeness is a deliberate commitment, not a shortfall, because GRUT's defining minimalism (one dynamical metric, one memory channel, Q) *forbids* the moves a derive-everything theory would need. The seams — dark matter HOSTED, the hierarchy magnitude and a second vacuum pole FORBIDDEN, the α -antecedent and the Born weights and the flavor mechanism OPEN, consciousness CONJECTURAL — are not the apology at the end of the story. They are the layered ontology showing through: here the universe makes X, here it must receive Y, here it forbids Z.

The honesty that lets the name stand is itself auditable. Every tier asserted across this document is checkable against `grut/toe/registry.py` — **121 claims across six machine-checked tiers (48 computed, 28 open_negative, 20 anchored, 11 conjectural, 10 meta, 4 foundational), which map onto the five reader-facing legend tiers as set out in the front matter** — and reproducible in `.venv`. A claim advances a tier only when a test certifies it; a failure is demoted and logged, never quietly dropped. The single failure mode that would sink the whole thing is a spine that quietly upgrades a HOSTED or OPEN result to DERIVED, and the apparatus is built to refuse that upgrade wherever it is tempting. So the word "Grand" is not earned by sweep, and "Responsive" is not a slogan: it is the literal floor of the ontology, the property $(Q \rightarrow F(t) \rightarrow \text{responsiveness})$ from which the vacuum, physics, complexity, and observation are each shown to descend.

Chapter 14 — The Decisive Test

There is one signal worth more than any single solved frontier, and it is the quiet one.

The strongest evidence that the ladder is real is **not** that every frontier is closed — most are not, and the framework says so out loud. It is that every derivation, every no-go, every hosted sector, and every audit — *including the failures* — keeps collapsing back onto the **same ladder**. The deepest instance is the unity of the two great no-gos: the dark-matter no-go and the hierarchy-magnitude no-go turn out to be one no-go, both forbidding the same thing — a new propagating vacuum pole — from the vacuum's own action, by locality and by Q. The same minimalism that *unifies* quantum mechanics and gravity as two exact limits of one action is the minimalism that makes those two unknowns structural rather than accidental. And so the 34-order hierarchy is not an embarrassment but a *protected* feature: it reduces to exactly two anchored numbers — $\ln(\tau_0/t_P) = 134.447$ and $\ln(\tau_{\text{micro}}/t_P) = 56.214$ — with the famous gap $c = 78.233$ being precisely their *difference*, not an independent third number (GRUT_HIERARCHY_LEDGER.md; verified to full precision in .venv). Existence forced, magnitude forbidden, size booked honestly as an input: the same fact placed at its true tier.

This is what makes the test *ongoing* rather than already-passed. The real question for GRUT-as-Theory-of-Everything is not "is it finished" — it is not, and it does not pretend to be; v4 is the current rung, and the destination is the same map with fewer OPEN cells. The real question is whether each *new* sector, when it is finally placed, finds its slot on this ladder — $Q \rightarrow F(t) \rightarrow \text{responsiveness} \rightarrow \text{vacuum} \rightarrow \text{physics} \rightarrow \text{complexity} \rightarrow \text{observation}$ — rather than demanding a piece of unrelated machinery bolted on from outside. So far the verdict has held where it has been put to the question. Dark matter, when pressed, resolved not into a new theory but into a resident of GRUT's own substrate sector $F(t)$ — internal to the layered ontology, costing free parameters, specified-and-unmet but in the right place. The hierarchy resolved into the two-anchor difference, protected by GRUT's own locality and Q rather than papered over. Locality, when interrogated, forbade the halo and forbade the pole with one theorem. The α -antecedent, even unsolved, declared its address precisely — IR-carrier identification, OPEN but attackable, not homeless. And Genesis, the hardest rung, placed the emergence of responsiveness itself on the ladder as a crossover, with Q surviving the loss of memory as the one invariant that carries through. Five hard cases; five times the same ladder; not once a demand for unrelated machinery.

That is the test, stated plainly and left running. Not omniscience — the framework's own minimalism forbids that, by design. The signal is the *recurrence*: that genesis, the vacuum, physics, the dark sector, the hierarchy, the observers, and even the things the framework cannot yet do all keep collapsing back onto one motion. A Theory of Everything, in the only sense that survives contact, is a complete tiered map with a sharp boundary — and it earns the name not when it has explained everything, but when everything it touches keeps finding the same place to stand. **That is when "Grand Responsive Universe Theory" stops being an ambition and becomes a description.**

Coda — The Universe's Current Model of Itself

Run the ladder all the way up and read what it spells.

The universe became responsive. Responsiveness produced stable structure. Stable structure produced observers. Observers produced models. Models produced GRUT.

GRUT is the universe's current model of its own responsiveness.

That one adjective — *current* — is the discipline of the whole book compressed to a single word. GRUT is not the universe's *final* account of itself; it is the best one its observers have built so far — a rung, not the arrival — written by responsive systems studying the responsiveness they are made of. The map will deepen: cells will move from OPEN toward DERIVED, or be booked honestly as hosted and forbidden; a deeper substrate layer may yet be detailed beneath the vacuum. But the shape of the thing is now visible — a universe that began as bare response, learned to remember, and, by way of physics, structure, and mind, turned back to recognize the responsiveness it had been all along.

That recognition is where the name is finally earned. Not because the theory explains everything — its own minimalism forbids that — but because it is the point at which a responsive universe became able to describe its own emergence.

Appendix A — Overclaim-Risk Table

The standing list of the claims most likely to be inflated by a careless reading, with the honest tier each must keep. These are the cells the spine refuses to upgrade.

#	Tempting overclaim	Honest standing
1	"GRUT derives / explains dark matter."	HOSTED. No cold DM is derived; the one forced relic is warm and refuted. DM is a resident of GRUT's own substrate $F(t)$ (2 dials, 0-of-3 sourced) — not external, not a successor theory.
2	"GRUT derives the hierarchy ($\tau_0 \gg \tau_{\text{micro}}$)."	Existence DERIVED; magnitude FORBIDDEN + anchored. The 34 orders reduce to two anchored Planck-logs whose difference is 78.233; no β -function carries τ_0 . The single spine-sinker if inflated.
3	"GRUT predicts $\Omega_\Lambda = 0.7145$."	Mechanism DERIVED; number anchored at 0.6886. The tree value $(2-R)^2 = 0.71453 \neq$ the full-Friedmann anchored 0.6886. Never conflate. The second spine-sinker.
4	"GRUT predicts $\eta_B = 6.6 \times 10^{-10}$."	Existence DERIVED; magnitude HOSTED-assisted. +7.7% over Planck, leaning on SM inputs (J_{CP} , K_{neq}). Not zero-parameter.
5	"GRUT explains consciousness / the universe is conscious."	CONJECTURAL hook only. A proven in-in self-comparison + a non-load-bearing 40 Hz coincidence. No account of experience; nothing depends on it.
6	" α is derived."	OPEN. The value $1/3$ is ADOPTED; only the conditional $a/c = 1/3$ math is proven; the IR-carrier antecedent is open.
9	"GRUT forces the Z_3 flavor structure."	HOSTED structure; identity DERIVED. The circulant is assumed; " $K = 2/3$ is a Z_3 identity" is the honest claim.
11	"GRUT predicts the neutrino ordering with no inputs."	DERIVED-from- $a_v=1$ (anchored). NH preference + $\Sigma m_v \approx 60$ meV ride on the $a_v = 1$ derivation, not on zero inputs.
13	"Enhanced growth / $\Omega_{dm} = \alpha$ (the v_2 dielectric line)."	FORBIDDEN + audit-trail-only. $\mu \neq 1$ refuted at $\sim 32\sigma$; recorded as dead, not resurrectable.
14	"GRUT derives the Born probabilities."	OPEN / HOSTED. Decoherence + pointer basis derived; the weights are received from the inner product. Not GRUT-specific.
15	" $f(T) = \tanh(T_c/2T)$ is a parameter-free theorem."	DERIVED [SPLIT]. A motivated identification (activation = zero-point fraction), trading a free width — not parameter-free.
16	"Gravity is derived as the elasticity of a solid vacuum."	G_0 value DERIVED; SOLID premise UNPAID POSTULATE. One of ≥ 4 degenerate medium classes.
17	"GRUT predicts A_s ."	OPEN (honest negative). No inflationary epoch; all three paths fail; A_s is anchored input.
18	"GRUT predicts n_s ."	CONJECTURAL (a fit). $n_s = 0.965$ obtained by solving for $H\tau$ from the observed value; the running is the genuine forward content.
19	"One action $\square$ GRUT explains everything."	A tiered PLACE for everything, not omniscience. Unification = the QM/GR limits + a complete map; gauge-coupling closure is OPEN.
20	"F (single-pole memory) is derived from Q."	Form DERIVED; pillar POSTULATED, τ_0 anchored. Single-pole-ness is not a theorem of Q alone (registry.py:5287).

#	Tempting overclaim	Honest standing
21	"The $\alpha = 1/3$ value is a GRUT theorem."	OPEN. Adopted axiom; only the conditional trace-anomaly ratio is proven.

Appendix B — Verified Numbers

All re-confirmed in .venv for this restructure. Where a tree value and an anchored value differ, both are listed so they are never conflated.

Quantity	Value	Note
α	1/3	Adopted dimensionless axiom (OPEN antecedent).
$R = \sqrt{4/3}$	1.15470	Deep-IR refractive index.
$(2 - R)^2$ (tree Ω_Λ)	0.71453	Tree-level; NOT the anchored 0.6886.
Ω_Λ (full-Friedmann, anchored)	0.6886	Planck 0.6889 (0.04%).
$S = 12\pi/\alpha^2 = 108\pi$	339.292	Screening factor.
$f(T_c) = \tanh(1/2)$	0.46212	Memory-amplitude order parameter at T_c ($\neq 1/2$).
$\ln(\tau_0/t_P) - \ln(\tau_{\text{micro}}/t_P)$	$134.447 - 56.214 = 78.233$	The hierarchy gap c is the <i>difference</i> of two anchors, not a third number.
$8\pi^2$	78.957	Rejected near-miss (+0.92%): universal one-instanton action, not GRUT-distinctive.
$16\pi^2$	157.91	Genuine $1/16\pi^2$ loop normalization (UV reference class).
$k_B T_c$	4.71 keV	Warm-relic energy (refuted cold-DM candidate).
$L_0 = c\tau_0$	12.85 Mpc	The one proper length breaking dilatation.
H_0	≈ 68.8 km/s/Mpc	In the Hubble-tension gap (anchored).
689 Hz	decoherence plateau	Gold microsphere, $1\ \mu\text{m}$, $m \approx 80.8$ pg; predicted test point (inherits τ_0).
$a_0 = cH_0/2\pi$	1.06×10^{-10} m/s ²	MOND-scale coincidence (not a derived MOND).
DM window center	$\log_{10}(m/\text{eV}) = -20.50$	Fuzzy-DM window $[-22, -19]$ middle.
DM geom-mean miss	7.19 dex	$\sqrt{E_{\text{micro}} \cdot E_{\tau_0}} = 4.84 \times 10^{-14}$ eV ($\log_{10} = -13.31$) vs center -20.50 .
τ -gap span	33.976 dex	$\log_{10}(\tau_0/\tau_{\text{micro}})$; partition $22.673 + 3.000 + 8.303 = 33.976$.
Registry census	121 = 48 computed / 28 open_negative / 20 anchored / 11 conjectural / 10 meta / 4 foundational	Machine-checked tiers.

Technical Appendix — The Load-Bearing Results

*This appendix exists to put the evidence in the referee's hands inside the document. The body runs the story; this gives the technical digest behind the seven results the whole map rests on — each with its key equations, its exact numbers, and its grounding in a specific `file:line` and registry tier in `grut/toe/registry.py`. It draws on the v4 banked audits: the core-chain skeleton, the hierarchy ledger, the UV-sector construction, the α -antecedent feasibility pass, the two dark-matter substrate tests, the $F(t)$ condensate program, and the four decisive-test docs. Every number below was re-confirmed in `.venv (numpy/sympy)`. The discipline is the same one the body holds to: a result is reported at its true tier, the split-tier seams are kept split, and nothing *HOSTED* or *OPEN* is dressed as *DERIVED*. Where a tree value and an anchored value differ, both are given so they are never conflated.*

A. The Core Derivation Chain — $\{Q, F, D, \alpha\} \rightarrow$ Constitutive Law $\rightarrow$ Results

GRUT's foundation is four inputs of unequal standing feeding one constitutive law, audited edge-by-edge as a skeleton with three tags: **SOLID** (genuine, zero hidden assumption), **COMPUTED** (real but parameter-dependent on α or τ_0 , premise named not hidden), **DOTTED** (a gap closed only by a premise the inputs do not supply). The audited skeleton is **3 SOLID, 3 COMPUTED, 5 DOTTED**, plus one Fourier step flagged as a definitional identity, not an edge.

The four inputs. Exactly one is a true zero-input theorem. **Q — the in-in causal arrow — DERIVED:** the Schwinger–Keldysh CTP action with the influence functional vanishing on the Keldysh diagonal, $S_{\text{IF}}[\varphi_+ = \varphi_-] = 0$; the closed–closed propagator identically zero; the retarded kernel vanishing for $t < 0$; FDT/KMS tying the noise to the dissipation with $N \geq 0$. Q is scale-free (no τ_0 , no τ_{micro} , no α), verified on a driven-oscillator example in all four legs, and is the only true zero-input SOLID node and the highest-fanout claim (93 downstream dependencies). (`ctp_action_structure`, *computed*, `registry.py:451`; `grut/foundation/ctp_action.py`, *verify 5/5*.) **F — finite single-pole memory — HOSTED [SPLIT]:** $\chi(\omega) = \alpha/(1 - i\omega\tau_0) \square \tau_0\dot{z} + z = z_{\text{target}} \square K(t) = (1/\tau_0)e^{-t/\tau_0}\Theta(t)$, one object in three forms. The single-pole *form* is the Mori–Zwanzig Markovian limit (derived, exact to $O(\tau_K/\tau_0)$); F-as-pillar and the *value* τ_0 are postulated/anchored — and the converse is false: single-pole-ness is *not* a theorem of Q alone, because passivity (Herglotz) is strictly weaker than complete monotonicity (a passive kernel can carry negative Debye weight or off-axis poles). (`finite_memory_form_from_q`, `registry.py:5278`, *false converse noted at :5295*; `first_order_from_mori_zwanzig`, `registry.py:5422`.) **D — broken adiabatic dilatation redundancy — OPEN [SPLIT]:** the one proper length $L_0 = c\tau_0$ breaks the boundary charge of the Weinberg $L_0 \rightarrow 0$ redundancy at $O((L_0 k)^2)$, non-anomalously (a diffeomorphism, measure Jacobian $\equiv 1$, so α does *not* enter); the breaking is derived, but the underlying redundancy is imported from Weinberg, with one residue (the scale-free in-in initial state, GAP-1, $n_s = 1$) open. (`adiabatic_dilatation_redundancy_nogo`, *computed*, `registry.py:5227`.) **$\alpha = 1/3$ — adopted dimensionless axiom — OPEN [SPLIT]:** sets the DC amplitude $\chi(0) = \alpha$, the index $R = \sqrt{1+\alpha}$, the screening $S = 12\pi/\alpha^2$; the conditional theorem (IF the gravitational conformal mode is the IR carrier of χ THEN $a/c = 1/3$, Komargodski–Schwimmer / Duff) is proven Fraction-exact, but the antecedent is open and the value adopted. (`alpha_vac_axiom`, *foundational*, `registry.py:238`; `alpha_vac_derivation`, *open_negative*, `registry.py:274`; `conformal_mode_scalar.py`, $a/c = \text{Fraction}(1,3)$, *9/9*.)

The hub. The four inputs assemble into one law, $\tau_0\dot{z} + z = z_{\text{target}}[z]$, every sector a regime of it. Q supplies the *form-class* (the CTP variation $\delta S_{\text{CTP}}/\delta z_a|_{\{z_a=0\}} = F[z_r]$ yields a causal, retarded, passive Herglotz relaxation toward a target, FDT fixing the noise — form-class only, no single pole); F upgrades it to the single-pole law via MZ, with τ_0 riding in as the anchor; D supplies the structural reading (it certifies $L_0 = c\tau_0$ is the controlled non-anomalous breaking and confines the nontrivial action to the **tensor** sector); α supplies only the normalization (the τ_0 dynamics is α -independent). The hub is therefore **DERIVED-as-synthesis [SPLIT]**. (`constitutive_equation`, *computed*, `registry.py:496`.)

The SOLID core — what survives with zero hidden assumptions. Four nodes are welded by derivation: **Q** itself; **E1** $Q \rightarrow$ hub form-class (causal + passive); **E8 $\mu_{\text{linear}} = 1$** , the transverse-tracefree projector theorem — P^{TT} annihilates the linear-scalar response, so linear cosmology is *exactly* Λ CDM, α -free and τ_0 -free, over-determined $\sim 32\sigma$ by the low- l CMB-ISW (the framework's cleanest theorem); and **E10** quantum mechanics recovered as the $\tau \rightarrow 0$ limit (one constitutive step is the first-order Euler–Schrödinger update, norm preserved,

$\langle \sigma_x \rangle = \cos \omega t$ on a precessing qubit; decoherence and a pointer basis recovered, the **Born weights NOT** — they are OPEN) together with the arrow of time $\dot{S} = (1/\tau_0)\langle (z - z_{\text{target}})^2 \rangle \geq 0$ (needs only $\tau_0 > 0$). Everything quantitative — R, S, Ω_Λ — is COMPUTED on the adopted α , not solid.

(adiabatic_dilatation_redundancy_nogo, registry.py:5227; qm_recovery, registry.py:753;
arrow_of_time_from_entropy, registry.py:1734.)

The minimal genuine input set: $\{Q, \alpha\}$ + the two anchors $\{\tau_0, \tau_{\text{micro}}\}$. Q is the proven backbone; α is the single load-bearing open axiom; τ_0 and τ_{micro} are the two anchored numbers, and their *difference* is the hierarchy ($\S B$), not a third number. F 's form and D 's breaking are derived *from these* (F via MZ given $\tau_{\text{micro}} \ll \tau_0$; D 's breaking given $L_0 = c\tau_0$). The core is not zero-input and does not pretend to be. (GRUT_V4_CORE_CHAIN.md §§2–6; zero_free_parameters *scopes the one-parameter claim to the gravitational core*, registry.py:404.)

B. The Two-Anchor Hierarchy Ledger — The 34-order Gap, Audited

GRUT's deepest unexplained fact is the ~ 34 -order separation between the macroscopic relaxation time $\tau_0 \approx 41.9$ Myr and the microscopic correlation time $\tau_{\text{micro}} \approx 1.40 \times 10^{-19}$ s. The ledger's result: **the whole hierarchy reduces to exactly TWO genuinely independent anchored dimensional numbers**, with everything else a disguise.

- $\tau_0 \approx 41.9$ Myr = 1.322×10^{15} s — dimensional anchor #1, anchored via H_0 /Bullet (two anchors agree ~ 10 – 20%); equivalently $\ln(\tau_0/t_P) = 134.45$ (verified 134.4472).
- $T_c = 54.7$ MK = 4.71 keV, with $\tau_{\text{micro}} \equiv \hbar/(k_B T_c)$ its SI-dual — dimensional anchor #2, anchored via the cosmic-chronology pin (T at $t \approx 16.5$ h post-BB); equivalently $\ln(\tau_{\text{micro}}/t_P) = 56.21$ (verified 56.2137).

The famous gap is $c = \ln(\tau_0/\tau_{\text{micro}}) = 78.233$, and it is **exactly the difference of the two independent Planck-logs**: $134.447 - 56.214 = 78.233$ (differenced before rounding, verified to machine precision) — **not** an independent third number. Every other hierarchy-flavored quantity is one of these two in costume: $L_0 = c\tau_0 = 12.85$ Mpc and $\ell_{\text{micro}} = c\tau_{\text{micro}}$ are unit conversions; the ratio $\tau_0/\tau_{\text{micro}} = 9.47 \times 10^{33}$ ($\log_{10} = 33.976$); $G_0 = \hbar/(c^3\tau_{\text{micro}}^4) \approx 1.03 \times 10^{16}$ Pa is the Debye relation written twice; $T_{\text{peak}} = \hbar/(\tau_0 k_B) = 5.78 \times 10^{-27}$ K is the τ_0 -dual temperature; $H_0 = 1/(S\tau_0)$ is *derived from* τ_0 , so H_0 is **not** an independent anchor. **$\alpha = 1/3$ is a separate dimensionless third number**, orthogonal to the τ -gap (it sets S , R , $n_g(0)$, never the 34-order scale).

Existence forced, magnitude unknown. The Mori–Zwanzig slow/fast split *forces* $\tau_0 \gg \tau_{\text{micro}}$ (it is the validity condition of the projection itself): the slow channel's existence and slow, gapless character follow from conservation ($\partial_\mu T^{\mu\nu} = 0$, the diffeomorphism Ward identity, part of Q), and the no-dark-mode result is verified magnitude-inert (ratios 10^{-1} through 10^{-34} all give the identical conclusion; the off-axis dark-pole criterion flips only at the pure ratio $\tau_K/\tau_0 = 1/4$). But the *size* is not forced: GRUT has no β -function carrying to — its only RG statement is that the τ_0 IR scale is RG-*protected* (the opposite of an asymptotic-freedom flow), so `relation_derivable = False` (Option B: two independently anchored scales). The tempting $8\pi^2 = 78.957$ ($+0.92\%$) is rejected by GRUT's own near-miss standard — it is the universal one-instanton action, not a GRUT-distinctive constant, with no derived tunneling linking the two sectors. The honest charge to any future derivation: force $\ln(\tau_{\text{micro}}/t_P) = 56.21$ *absolutely* (against t_P) from a real condensation coefficient with τ_0 never entering; deriving c directly is circular (c already contains $\log \#1$). — **existence FORCED; magnitude FORBIDDEN-to-derive + anchored.** (GRUT_HIERARCHY_LEDGER.md §§2–5; tau_hierarchy_decision.py:358 `relation_derivable=False`; mori_zwanzig_kernel.py:99–105.)

C. The Unified No-Go — Dark-Matter No-Go = Hierarchy No-Go

The deepest structural fact the skeleton exposes (edge E9): **the no-go that hosts dark matter and the no-go that forbids the hierarchy magnitude are the SAME no-go.** A *derived* cold dark sector and a β -function carrying τ_0 would each require the *same* object — a new propagating vacuum pole — and that object is forbidden from the vacuum's own action by two gates that are really one.

- **The locality / no-halo theorem.** A GRUT dark-sector response that is covariant, local in the matter fields (kernel analytic — entire, pole-free — in k^2 near $k = 0$) and nonlinear is no more spatially extended than its baryon source; an extended $\rho \propto 1/r^2$ halo would require the response to be singular at $k = 0$ (a $1/k^2$ inverse-Laplacian, $g = \nabla(1/\nabla^2)\rho$), which locality forbids. Hence a derived dark sector requires a *new pole in the vacuum spectrum — a new propagating mode, not a new operator*. GRUT's actual kernel $\sigma \cdot \alpha \cdot \hat{\chi}(\omega) \cdot P^{\text{TT}}$ is on the permitted (analytic) side. (`locality_no_halo_theorem`, `registry.py:5540`; `locality_no_halo.py:7–14`.)
- **The propagating-relic pincer (Ostrogradsky = Q).** A propagating vacuum pole needs a higher-derivative TT operator $\square$ an Ostrogradsky ghost $\square$ opposite-sign residues ($+1/m^2, -1/m^2$, verified `apart(1/(s(s+m^2)))`) $\square \text{Im } \chi < 0$ $\square$ by FDT the noise $N = 2\hbar \coth(\hbar\omega/2kT) \cdot \text{Im } \chi < 0$ $\square$ the off-diagonal influence weight $1/2 h_a N h_a$ is not positive-semidefinite $\square$ the in-in path integral is non-unitary.
Ghost-freedom and Q are the same gate. (`propagating_relic_forbidden_pincer`, *conjectural*, `registry.py:5479`.)

The α -antecedent structural-NO and the locality–dynamics fork. The same pincer closes the UV/QG frontier and the dynamical completion of the α -antecedent. A UV-dynamical sector that would force $\ln(\tau_{\text{micro}}/t_P)$ needs a live running coupling; GRUT's only GRUT-origin coupling is the trace-anomaly coefficient, which is one-loop-exact ($\beta = 0$) and lives in the **E_4/a -anomaly channel that GRUT's own second-order analysis proves is Lovelock-null / dynamically dormant in 4D**, while the live channel is c/W^2 (TT) — *not* where the Riegert/conformal-mode action is built. To do dynamical work the conformal mode σ must couple to the matter trace (it is the metric trace, tied to T by the Ward identity), producing $\sigma(k) = T/(Q^2 k^4)$: a **$1/k^4$ matter→metric pole, strictly worse than the forbidden $1/k^2$** (verified $\lim_{k \rightarrow 0} \{1/k^4\}/\{1/k^2\} = \infty$), and a 4th-order Paneitz ghost = Q-violation. This is the **locality–dynamics fork: locality and Q are saved only by killing the dynamics; doing dynamical work trips locality and Q**. There is **no window** where σ is simultaneously locality-safe, ghost-free, and dynamically live. The α -value's antecedent therefore resolves to **kinematic identification = NOT-YET-FORMULATED** (candidate-selected by monopole dominance, self-demoted to necessary-not-sufficient; the one outstanding piece is the 4th-order S^4 Riegert/Paneitz a/c ratio, never computed) + **dynamical completion = STRUCTURALLY-BLOCKED** (it inherits the UV no-go). The conditional theorem stands; α remains the adopted axiom — the verdict forecloses moving α *up* to established, and does not move it *down*. The only escape from the unified no-go — a new propagating vacuum pole with its own un-GRUT-sourced running coupling — *fractures GRUT's minimalism: it constitutes a different theory, not GRUT.* — **FORBIDDEN-BY-THEOREM.** (`GRUT_UV_SECTOR_CONSTRUCTION.md §§4–5`; `GRUT_ALPHA_ANTECEDENT_FEASIBILITY.md §§2,6`; `second_order_kernel.py:16–22`.)

D. Dark Matter as Substrate-Resident — The F(t) Condensate Specification

Run the relic inventory forward without biasing it toward producing dark matter and **no viable cold DM falls out**. The single-mode structure forbids a second stable *relaxational* pole (a channel-counting postulate, not a consequence of Q alone); the one condensation-tied relic GRUT can force — an elastic shear phonon at $k_B T_c = 4.71$ keV — is **warm**, travels at $c_s = c$, and overcloses the universe by $\sim 418 \times -5800 \times$. GRUT does not predict *no* dark matter; it predicts the *wrong* dark matter, and books that as a genuine negative. **DM is HOSTED**.

(GRUT_GENESIS.md §4; locality_no_halo_theorem, computed.)

The two DM substrate tests then sharpen "hosted" into **specified-and-unmet to the dex**. The three no-gos, read together, are a 4-item spec for the missing host: (1) non-thermal / coherent (zero entropy, $\langle \phi \rangle \neq 0$); (2) an *intermediate* mass in the empty middle of the 34-order τ -gap; (3) sourced *outside the vacuum action* — in the pre-responsive substrate **F(t)**, beneath the vacuum, where the no-gos have no jurisdiction (they are theorems about the *vacuum's* action); (4) gravitates by ordinary GR, clusters, stable ≥ 14 Gyr, delivers $\Omega_{dm} h^2 \approx 0.12$. The empty middle is mapped exactly: the window $\log_{10}(m/\text{eV}) \in [-22, -19]$ (center -20.50) sits **22.673 dex below** $\hbar/\tau_{\text{micro}} = 4713.68$ eV and **8.303 dex above** $\hbar/\tau_0 = 4.978 \times 10^{-31}$ eV; the partition closes to the dex ($22.673 + 3.000 + 8.303 = 33.976$). The natural dimensionless bridge — the geometric mean $\sqrt{(E_{\text{micro}} \cdot E_{\tau_0})} = 4.844 \times 10^{-14}$ eV ($\log_{10} = -13.31$) — **misses the window center by 7.19 dex**, so a new scale is genuinely needed, not disguisable as a derived one.

The minimal construction is a textbook misalignment scalar placed in F(t): $V(\phi) = \frac{1}{2}m^2\phi^2$, EOM $\phi'' + 3H\dot{\phi} + m^2\phi = 0$, frozen at ϕ_i until $H \sim m$ then oscillating and redshifting as a^{-3} (cold dust, automatically stable $\gg 14$ Gyr; verified $m/H_0 \approx 6.9 \times 10^{10} \gg 1$ at $m = 10^{-22}$ eV), gravitating via ordinary $T_{\mu\nu} = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}m^2\phi^2$ through the permitted Newtonian Green's function, with abundance $\Omega_{dm} h^2 \approx 0.1 \cdot (\phi_i/10^{17} \text{ GeV})^2 \cdot (m/10^{-22} \text{ eV})^{0.5}$.

Honest cost ledger: exactly 2 inserted dimensionful dials — the mass $m \in [10^{-22}, 10^{-19}]$ eV (a genuine third frequency) and the amplitude $\phi_i \sim 10^{16-17}$ GeV — plus 1 inserted potential shape; 0 of 3 requirements sourced from the bath. The bath supplies only two frequencies ($1/\tau_0, 1/\tau_{\text{micro}}$), a dissipative noise spectrum $S_h(\omega, T=0) \propto \omega/(1+(\omega\tau_0)^2)$ with $\langle h \rangle = 0$ (it *fluctuates*, it does not *condense*), and the dimensionless order parameter $f(T_c) = \tanh(1/2) = 0.4621$ — none of which carries the required mass, VEV, or GeV amplitude. It is **no-go-clean** (ordinary $T_{\mu\nu}$, ordinary collapse, no $1/k^2$, no second vacuum pole, no new relaxational mode) and **Q/KMS-clean** ($N \geq 0$, no $\text{Im } \chi < 0$). It is **GRUT detailing its own substrate sector — a CONJECTURAL deeper layer, not a different theory and not a successor** (the "different-theory" verdict belongs to the *vacuum-pole* route, not to this substrate detailing) — but it is **not a v4-core derivation**, because it adds 2 free parameters. The analogy: F(t)-DM is to GRUT what the Standard Model is to GR's $T_{\mu\nu}$.

Falsifier: a 10^{-22} eV scalar makes kpc-scale soliton cores ($\lambda_{dB} = 1.917$ kpc at $v = 10$ km/s) and suppresses small-scale structure; the Lyman- α forest already pushes the floor toward $\sim 2 \times 10^{-21}$ eV, and if observations force $m \geq 10^{-19}$ eV the kpc-core window closes and the misalignment host loses its phenomenology.

(GRUT_DM_NONTHERMAL_COHERENT_TEST.md; GRUT_FT_CONDENSATE_PROGRAM.md; registry.py:5340/:5471/:5492/:5532.)

E. The Decisive-Test Scorecard — Four Verdicts and an Untested Seam

For each Part IV frontier the decisive question is not "is it solved" but *does the sector organize around responsiveness — finding its slot on the ladder $Q \rightarrow F(t) \rightarrow \text{Responsiveness} \rightarrow \text{Vacuum} \rightarrow \text{Physics} \rightarrow \text{Complexity} \rightarrow \text{Observation}$ — or does it require unrelated machinery bolted on from outside?* A container hosts a result (right numbers, structure adopted from elsewhere); an *explanation* generates the structure from the spine. Each verdict was allowed to come back negative.

- **Dark matter — ORGANIZED (substrate-resident).** Pressed, dark matter resolves not into a new theory but into a no-go-clean coherent scalar resident in GRUT's own substrate sector $F(t)$ — internal to the layered ontology, costing free parameters, specified-and-unmet but in the right place (§D). The *make-or-break*: whether the bath can source the host. It cannot (0 of 3) — but the host lives inside GRUT's existing substrate slot, not in a successor.
- **Quantum gravity / gravitation — PARTIAL (leaning ORGANIZED).** The two legs split cleanly. **Gravity-as-response is DERIVED:** $\Phi_{\mu\nu} = \alpha \cdot \hat{\chi}(\omega) \cdot P^{\text{TT}}$ from varying the constitutive cross term $S_{\text{const}} = -\frac{1}{2} h_a(\alpha \cdot \hat{\chi}) h_r$, with $\hat{\chi}(\omega) = 1/(1-i\omega\tau_0)$ the normalized response pole (the full susceptibility is $\chi = \alpha \cdot \hat{\chi} = \alpha/(1-i\omega\tau_0)$); $\mu_{\text{linear}} = 1$ as the P^{TT} projector theorem (over-determined $\sim 32\sigma$ by ISW); GR recovered as the high- ω / DC limits ($n_g(0) = \sqrt{4/3} = 1.1547005$, $n_g(\infty) = 1$). **The UV frontier is a framework-internal no-go (§C).** **But gravity-as-geometry is HOSTED:** the linearized Einstein tensor enters as a literal placeholder symbol `G_einstein_one = sp.Symbol("G_einstein_1")`, the EoM read off is $G_{\mu\nu}^{(1)} - \Phi_{\mu\nu} = 8\pi G T_{\mu\nu}$ with $G_{\mu\nu}^{(1)}$ on the *input* side, the Einstein–Hilbert term is one of four input pieces, and the metric $g_{\mu\nu} = \eta + h$ is a convention. GRUT derives the *response of gravity*, not the *geometry of gravity*. The would-be deeper account (the elastic-medium/ G_0 picture) is demoted as an unpaid postulate that predicts a refuted warm dark sector. The *make-or-break*: a paid, registry-tier derivation of the geometric structure from the constitutive law. (`phi_munu_linearized_derivation`, `registry.py:3719`; `gr_recovery`, `registry.py:1071`; `GRUT_QG_RESPONSIVENESS_TEST.md`.)
- **Neutrinos / flavor — CONTAINER-SEAM.** The *consequences* are real and machine-verified (Koide $K = 2/3$ to machine precision at $a = \sqrt{2}$, $N = 3$, the normal-ordering preference with $\Sigma m_\nu = 59.6$ meV conditional on $a_\nu = 1$, the sharp non-extension of the charged-lepton Z_3 to neutrinos at 194.1 vs observed 33.9), but the *structure that produces them* — the Z_3 circulant $\sqrt{m\Box} = M_0(1 + a \cdot \cos(\theta + 2\pi k/3))$, the amplitude $a = \sqrt{2}$, the scale M_0 — is an adopted Yukawa ansatz GRUT hosts and does not generate. The responsive route was *tested* and returns the wrong number: GRUT's impedance balance $2\rho = \alpha_{\text{vac}} = 1/3$ gives $K = 4/9$, not $2/3$ (a $\sim 50\%$ miss; the Koide amplitude sits at $4.24 \times \alpha_{\text{vac}}$). M_0 is a free fit ~ 20 orders from the native $\mu_0 = \hbar/\tau_0$. The one tie to the spine, $\theta = 2/9 = K \cdot \alpha_{\text{vac}}$, is statistically real (4.62 ppm, $557\times$ margin) but mechanistically decorative and collapses to a single α_{vac} via $K \equiv 2 \cdot \alpha_{\text{vac}}$. The seam points *away* from the ladder. **The make-or-break:** a derivation of the Z_3 -circulant Jacobian $+a = \sqrt{2}$ from the CTP fixed point, or a foundation-anchored M_0 . **Falsifier:** a measured inverted ordering at $>5\sigma$. (`koide_z3_circulant_structure`, `registry.py:2808`; `neutrino_hierarchy_z3_nh_prediction`, `registry.py:2926`; `GRUT_NEUTRINO_RESPONSIVENESS_TEST.md`.)
- **Baryogenesis — CONTAINER-SEAM (weaker), with the overclaim corrected.** $\eta_B = J_{\text{CP}} \times K_{\text{neq}} \times (2-R_B)/S_B = 6.57 \times 10^{-10}$ ($+7.7\%$ vs Planck 6.10×10^{-10}) is real, but every decisive check fails: $(2-R_B)/S_B$ is **not** the canonical α -spine ($R_B = 1.018 \neq R = \sqrt{4/3} = 1.155$; $S_B = 4\pi \times 45 = 565.5 \neq S = 108\pi = 339.3$); the GRUT factor is **not load-bearing** (its numerator $(2-R_B) = 0.982$ is $O(1)$ and the full factor $\approx 1.7 \times 10^{-3}$ moves η_B by $<2\%$ — the doc itself says the smallness comes from $J_{\text{CP}} \times K_{\text{neq}}$); the

magnitude is **hosted/imported** (J_CP is SM/CKM, K_neq is a hardcoded $\pm 50\%$ uncomputed non-equilibrium number); the $+7.7\%$ was **produced by a documented S_B re-choice**; and using the *canonical* spine R *worsens* the fit ($+7.7\% \rightarrow -7.3\%$). This is the **baryogenesis overclaim correction**: the earlier "GRUT predicts η_B " reading is corrected to existence DERIVED / magnitude HOSTED-assisted, a reverse-fit booked honestly. The one genuinely GRUT-flavored idea (the constitutive lag supplying non-equilibrium across the EW crossover) is on-spine as a *narrative*; the *number* is not. **The make-or-break**: derive K_neq from the lag in code *and* show the canonical R/S still land η_B (fails by construction today). (baryogenesis_eta_b, *anchored*, registry.py:1696; GRUT_BARYOGENESIS_RESPONSIVENESS_TEST.md.)

- **Coupling unification — UNTESTED through the lens (no verdict asserted; likely CONTAINER-SEAM, OPEN).** The three SM gauge couplings, their β -functions, and the 8.9% unification miss are 100% hosted SM (reproduced in .venv from PDG couplings + Machacek–Vaughn 1-loop β with zero GRUT input: closest-approach scale $10^{14.37}$ GeV, miss 8.97%). The responsive correction $\Delta\beta(\alpha_{\text{eff}}(\omega))$ is a prose conjecture — a *named* hook (memory kernel $K(t) \rightarrow$ non-Markovian $\delta\beta_i$) but no equation, no tie to $\chi(\omega)$ in the gauge sector, no derivation from $\delta S_{\text{CTP}}/\delta g_i$, no numerical closure (zero lines of code). Distinct from the other two seams: this one was **never attempted** (OPEN/unbuilt), not tried-and-missed; and it carries no overclaim — open_negative is already the honest tier.
(track_v_coupling_unification_open_question, *open_negative*, registry.py:2718; GRUT_COUPLING_UNIFICATION_RESPONSIVENESS_TEST.md.)

Scorecard. Category A (on-ladder): **Dark Energy ORGANIZED**, **Hierarchy ORGANIZED** (existence), **α ORGANIZED** (value-adopted), **Dark Matter ORGANIZED** (substrate-resident in $F(t)$), **Quantum Gravity PARTIAL** (response derived, geometry hosted). Category B (off-ladder, hosted SM matter-content): **Neutrinos**, **Baryogenesis — CONTAINER-SEAM** (Coupling Unification untested through the lens — likely CONTAINER-SEAM, no verdict asserted). Category C (observer-meta): **Consciousness / Information / Mathematics — conjectural/open**. The pattern holds: vacuum/gravitational sectors are on the ladder; gauge/matter-content sectors are off it.

F. The Falsifier Table — The Sharp Edges

Each row is a single, attackable prediction whose failure would demote the claim it carries.

Falsifier	Prediction	Demotes / kills	Tier
$\mu_{\text{linear}} = 1$	Linear cosmology is exactly Λ CDM; a $\mu \rightarrow 4/3$ scalar enhancement gives a $2.79\times$ ISW excess, refuted at $\sim 32\sigma$ by the low- ℓ CMB-ISW. A detected linear $\mu \neq 1$ breaks the $P^{\wedge}TT$ projector theorem.	the response-derivation / E8	DERIVED
689 Hz decoherence plateau	A gold microsphere ($1\text{ }\mu\text{m}$, $m \approx 80.8\text{ pg}$) shows a decoherence plateau at 689 Hz, inheriting τ_0 . A null at the predicted point falsifies the τ_0 -tied decoherence prediction.	the gold-benchmark decoherence point	DERIVED (inherits τ_0)
Normal neutrino ordering	Normal ordering preferred, $\Sigma m_\nu \approx 60\text{ meV}$ ($m = 0.8, 8.7, 50.2\text{ meV}$), conditional on $a_\nu = 1$. A measured inverted ordering at $>5\sigma$ falsifies it.	the NH export + the $a_\nu = 1$ chain	DERIVED-from-anchored
Wave-DM soliton scale	If DM is the $F(t)$ misalignment host ($\sim 10^{-22}\text{ eV}$), it makes kpc-scale soliton cores ($\lambda_{\text{dB}} = 1.917\text{ kpc}$ at $v = 10\text{ km/s}$). Lyman- α + dwarf-satellite counts forcing $m \gtrsim 10^{-19}\text{ eV}$ close the window.	the $F(t)$ substrate host	CONJECTURAL
η_{B} band	$\eta_{\text{B}} \approx 6.57 \times 10^{-10}$; a refined PDG/Planck value outside the $\sim 8\%$ band falsifies the export. (CMB-S4 resolves $+7.7\%$ at $>20\sigma$ — a deviation <i>falsifies</i> , it does not vindicate.)	the baryogenesis export	CONTAINER-SEAM
Coupling unification	High-precision collider running showing the couplings do not unify at any scale falsifies the gauge-structural prediction.	the coupling-unification conjecture	OPEN
$8\pi^2$ rejected	$8\pi^2 = 78.957$ ($+0.92\%$ over $c = 78.233$) is rejected by GRUT's own near-miss standard — the universal one-instanton action, not a GRUT-distinctive constant.	guards against numerology	—

Anchored cosmology numbers carried alongside, never conflated: $\Omega_\Lambda = \mathbf{0.6886}$ (full-Friedmann anchored; Planck 0.6889, 0.04% match) versus the **tree** $(2-R)^2 = \mathbf{0.71453}$ (cosmic-baseline approximation); $R = \sqrt{4/3} = 1.15470$; $S = 12\pi/\alpha^2 = 108\pi = 339.292$; $H_0 \approx 68.8\text{ km/s/Mpc}$.

G. The Machine-Checked Apparatus — 121 Claims, Six Tiers, Five-Tier Legend

The honesty that lets the name stand is itself auditable. Every tier asserted in this document is checkable against `grut/toe/registry.py` and reproducible in `.venv`. A claim advances a tier only when a test certifies it; a failed result is demoted and logged, never quietly dropped (over 3,000 gating tests). The single failure mode that would sink the whole thing — a spine that quietly upgrades a HOSTED or OPEN result to DERIVED — is the one the apparatus is built to refuse, and the meta-layer that performs that audit is itself DERIVED.

The six machine-checked registry tiers and census (121 claims):

Registry tier	Count	Meaning
computed	48	A real derivation (parameter-dependent where it rides α or τ_0).
open_negative	28	No mechanism yet, attackable; honest negatives booked open.
anchored	20	Fixed to observation (τ_0 , T_c , the Ω_Λ value, ...).
conjectural	11	A structural hook only, far from derivation.
meta	10	The audit layer — the registry checking itself.
foundational	4	Adopted axioms ($\alpha = 1/3$ by adoption, ...).

The five-tier reader legend → six machine tiers map. The narrative carries five reader-facing tags; they relate to the six machine tiers by one map (not two competing systems): **DERIVED** $\approx$ computed or foundational; **HOSTED** $\approx$ anchored; **OPEN** $\approx$ open_negative; **CONJECTURAL** $\approx$ conjectural; **FORBIDDEN-BY-THEOREM** is a narrative reading of computed/conjectural no-go claims; and **meta** is the audit layer, carrying no reader-facing tier of its own. Several events are **split-tier seams** ([SPLIT]): the mechanism is DERIVED but its input anchored, or a *form* is DERIVED while its *antecedent* is OPEN — and the split is the honest content; it must never be collapsed to the higher tier. This is the apparatus behind every tier in the body, and behind the appendix's refusal to upgrade a single dotted edge. (GRUT_V4_SPINE.md "*How to read this*"; the registry census re-verified in `.venv`.)